

Secure Control Module

PROJECT CODE: BRDZ1C010001

PCB NUMBER: 21X000

PCB VERSION: SA

PCB SIZE: 120.4 x 90 mm

PCB Layer: 8 Layer

PCB THICKNESS: 1.57 ±10% mm

<Variant Name>

		Wiwynn Incorporated 8F, No.90, Xintai 5th Rd., Xizhi Dist, New Taipei, Taiwan http://www.wiwynn.com	
Title COVER PAGE			
Size A3	Document Number 21X000		Rev SA
Date:	Monday, September 27, 2021		Sheet 1 of 67

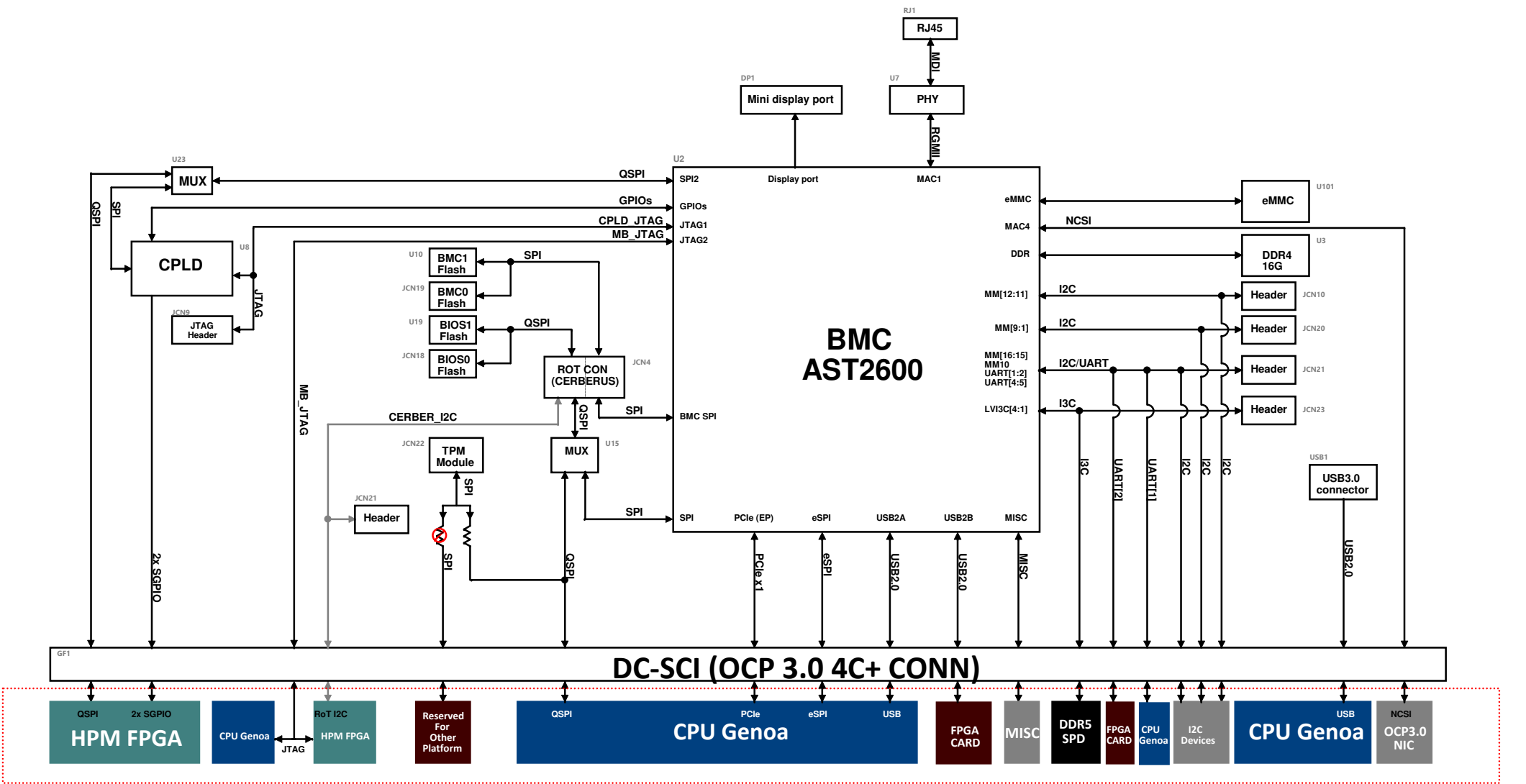
INDEX

PAGE	COMPONENT / FUNCTION	PAGE	COMPONENT / FUNCTION	PAGE	COMPONENT / FUNCTION
001	COVER_PAGE	026	BMC_MEMORY	051	BLANK
002	INDEX	027	BMC_DDR	052	FUNCTION_LED/FRU
003	BLOCK_DIAGRAM (AMD)	028	BMC_UART	053	BLANK
004	BOOT_UP_SEQUENCE_DIAGRAM (AMD)	029	BMC_MAC	054	I2C_I3C_UART_HEADER
005	POWER_ON_SEQUENCE (AMD)	030	BMC_ADC	055	PLD_JTAG_HEADERS
006	POWER_OFF_SEQUENCE (AMD)	031	BMC_DISPLAY_PORT	056	BLANK
007	POWER_DELIVER_DIAGRAM	032	BMC_POWER	057	BLANK
008	CLOCK_BLOCK_DIAGRAM (AMD)	033	BMC_POWER_CAP	058	BLANK
009	I2C_BLCOK_DIAGRAM_BMC/PCH (AMD)	034	BMC_GND	059	P5V_STBY
010	SGPIO_BLOCK_DIAGRAM	035	BMC_STRAPS	060	P3V3_SCM_STBY
011	SPI_BLCOK_DIAGRAM (AMD)	036	BMC_RGMII_PHY	061	P2V5_BMC_STBY
012	CERBERUS_BLOCK_DIAGRAM (AMD)	037	BMC_eMMC	062	P1V2_BMC_STBY
013	CERBERUS_CS_BLOCK_DIAGRAM_BMC	038	CPLD (1/3)	063	P1V8_BMC_STBY
014	THERMTRIP_&_PWRBRK#_DIAGRAM(AMD)	039	CPLD (2/3)	064	P1V0_BMC_STBY
015	SYSTEM_ERROR_REPORTING (AMD)	040	CPLD (3/3)	065	P0V6_DDR_VREF_STBY
016	MEMORY_WARNING_REPORTING	041	TPM_CONNECTOR	066	MECHANICAL_HOLE_COUPON
017	PROCHOT_DIAGRAM (AMD)	042	BIOS_FALSH	067	HISTORY
018	SW_NVDIMM_BLOCK_DIAGRAM	043	CERBERUS_CONNECTOR	068	
019	UART_BLOCK_DIAGRAM	044	BLANK	069	
020	JTAG_BLOCK_DIAGRAM	045	BMC_FLASH	070	
021	HDT_BLOCK_DIAGRAM (AMD)	046	HPM_CPLD_SPI_MUX	071	
022	STACKUP	047	USB	072	
023	SCM_JUMPER_SETTING_LED_DEFINE	048	TEMPERATURE_SENSORS	073	
024	DC-SCI	049	BLANK	074	
025	BLANK	050	POWER_RESET_BUTTON_LOGIC		

<Variant Name>

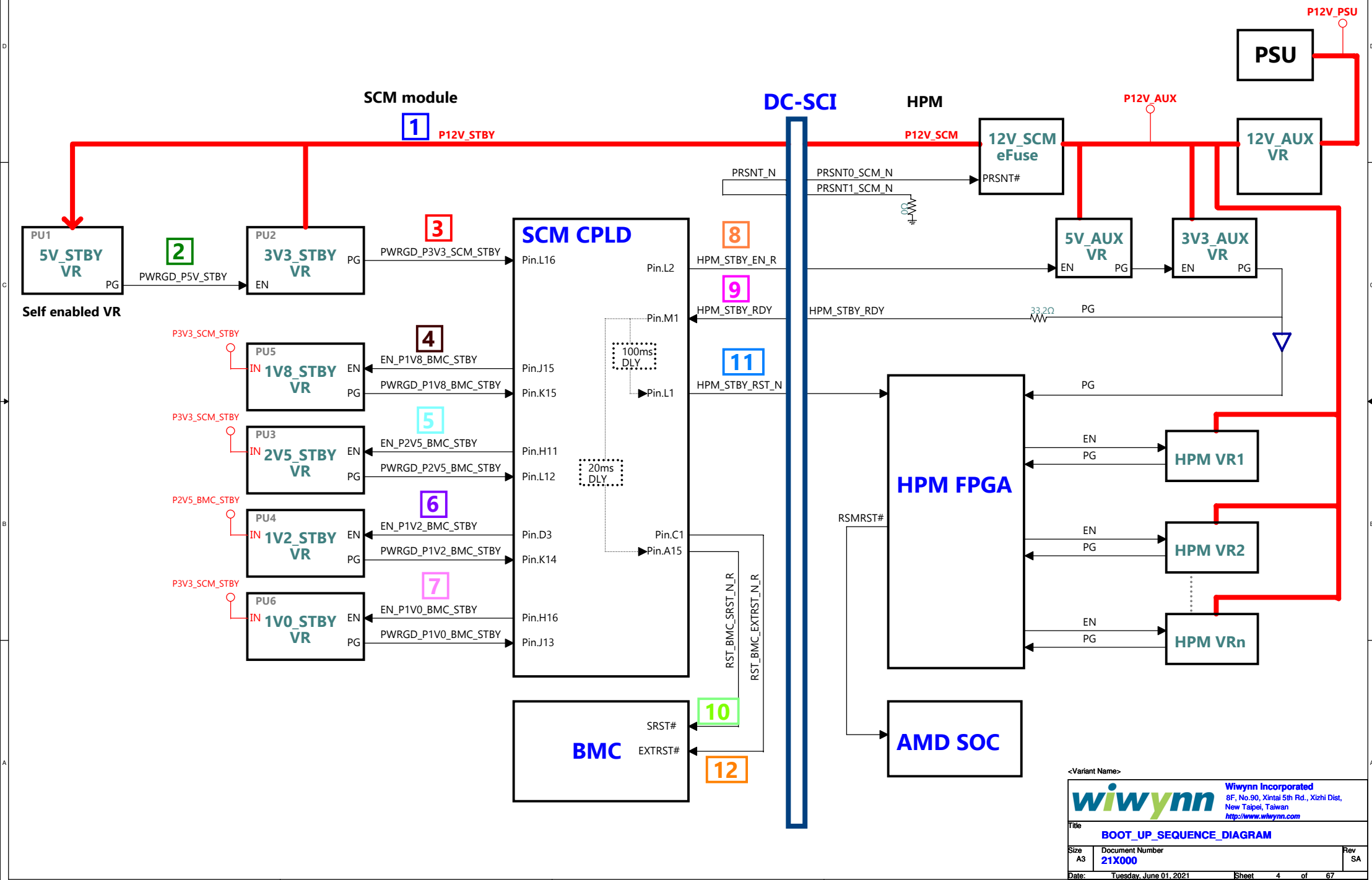
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Title INDEX			
Size A3	Document Number 21X000		Rev SA
Date:	Tuesday, June 01, 2021	Sheet	2 of 67

BLOCK DIAGRAM (AMD)



Based on C2195 HPM architecture.

BOOT_UP_SEQUENCE_DIAGRAM (AMD)



POWER_ON_SEQUENCE

Sequence on HPM side
Sequence on SCM side

HPM Side

P12V_PSU
P12V_AUX
P12V_SCM (To SCM)

SCM Side

P12V_STBY (From HPM)
P5V_STBY
P3V3_SCM_STBY
P1V8_BMC_STBY
P3V3_SCM_RGM_STBY
P2V5_BMC_STBY
P1V2_BMC_STBY
P1V0_BMC_STBY
HPM_STBY_EN (To HPM)

HPM Side

HPM_STBY_EN (From SCM)
P5V_STBY
P3V3_STBY
⋮
HPM_STBY_RDY (To SCM)

SCM Side

HPM_STBY_RDY (From HPM)
RST_BMC_SRST_N
HPM_STBY_RST_N
PORT0_RST_N
RST_BMC_EXTRST_N
PORT1_RST_N (To HPM)

HPM Side

PORT1_RST_N (From SCM)
P0_RSMRST_N
⋮
P0_PCIE_RST0_N

POWER_OFF_SEQUENCE

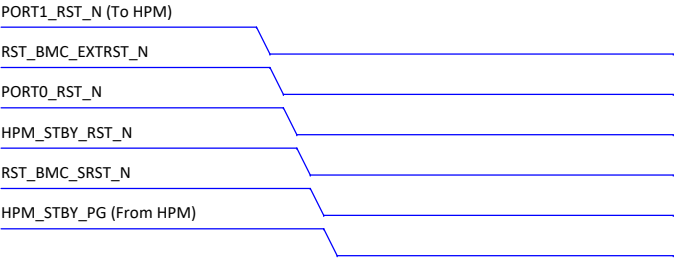
Sequence on HPM side
Sequence on SCM side

TBD

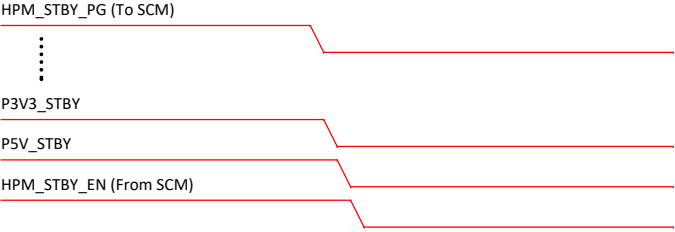
HPM Side



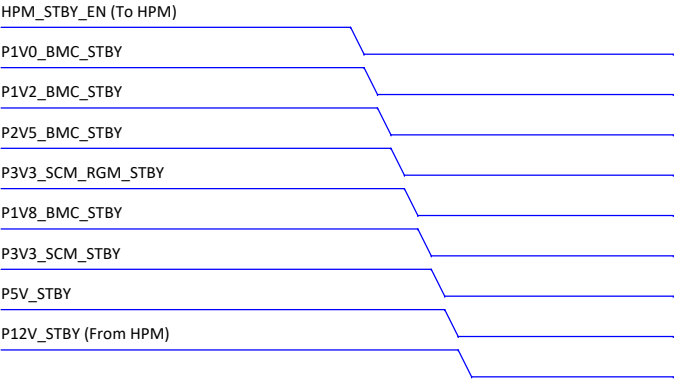
SCM Side



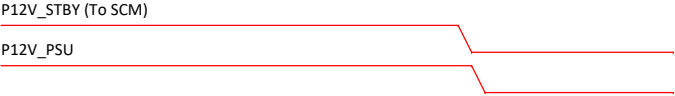
HPM Side



SCM Side

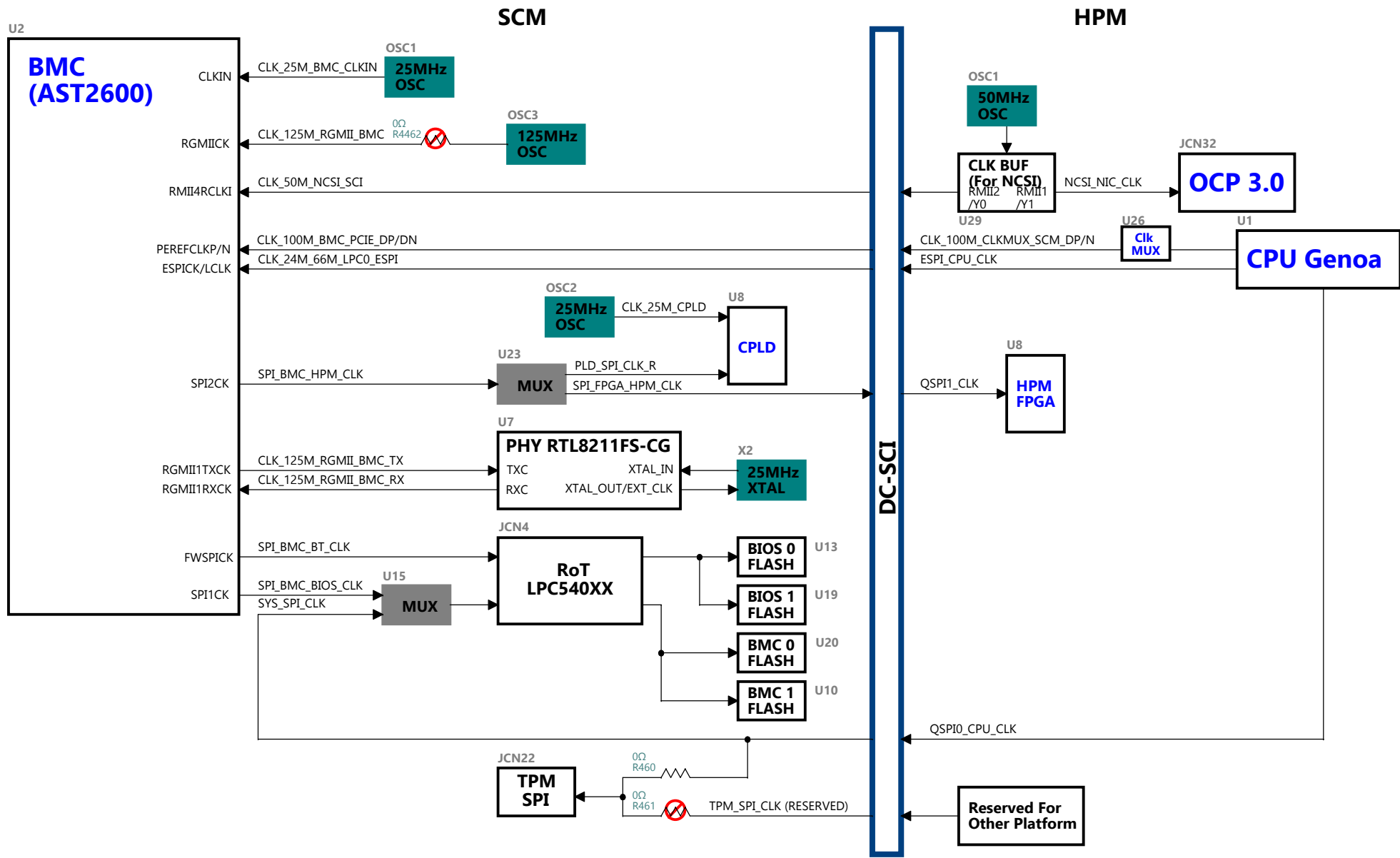


HPM Side

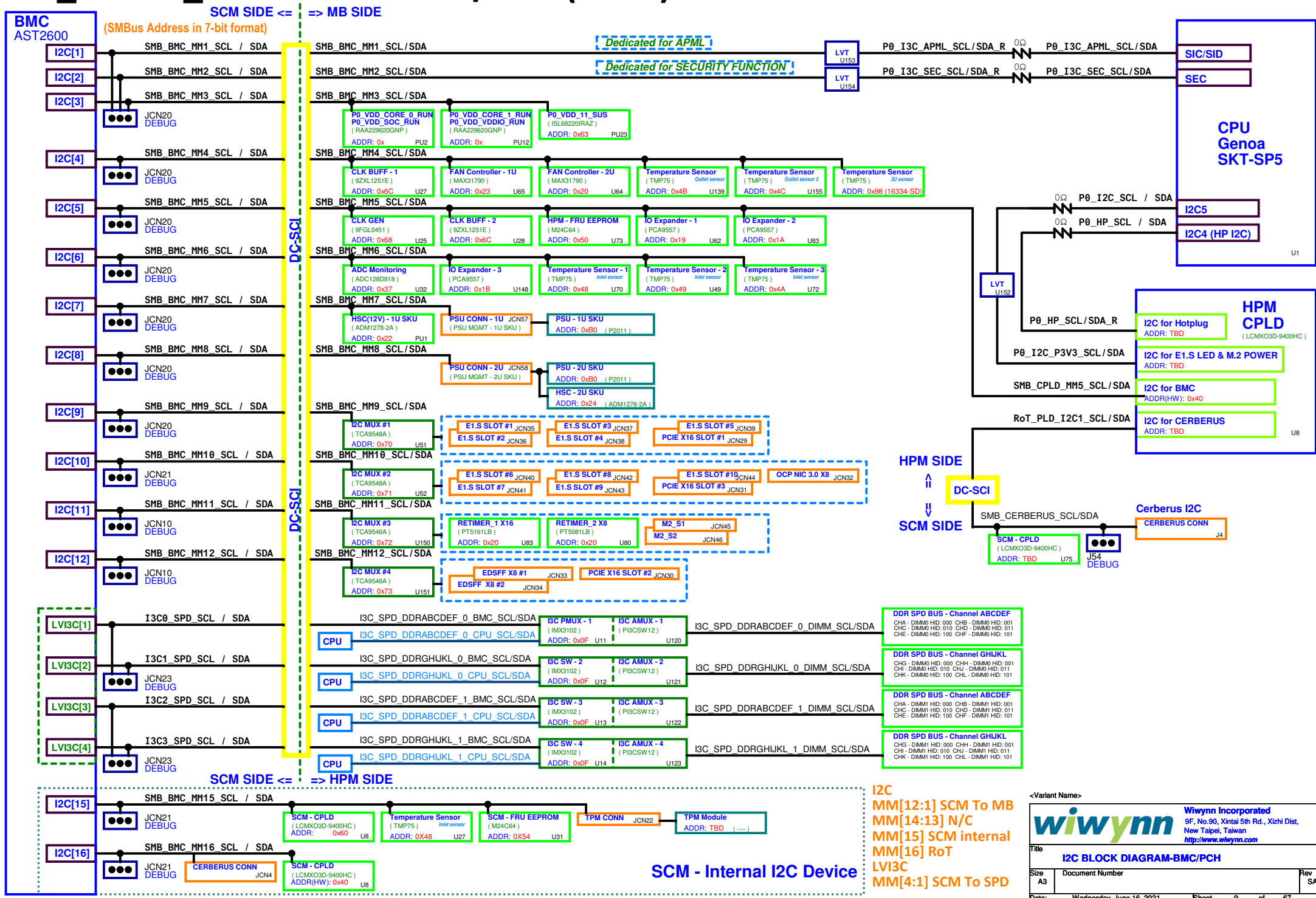


The diagram illustrates the power distribution for the DC-SCI block. The DC-SCI input is connected to a network of power planes and regulators. The power planes shown are P3V3_SCM, P12V_STBY, and P5V_STBY. The regulators shown are P3V3_SCM_STBY (Max: 5.42A), P1V8_BMC_STBY (Max: 0.63A), P1V0_BMC_STBY (Max: 0.71A), P2V5_BMC_STBY (Max: 1.58A), P1V2_BMC_STBY (Max: 1.28A), and P0V6_DDR_VREF_STBY (Max: 0.33A). The regulators are connected to various components, including BMC AST2600, BMC DRAM DDR4, CPLD, SPI Flash x4, PHY RTL8211FS-CG, TPM SPI, CERBERUS, DISPLAY PORT, USB3.0 CONN, eMMC, and MISC. The diagram also shows the connection of the DC-SCI input to the P3V3_SCM and P12V_STBY planes.

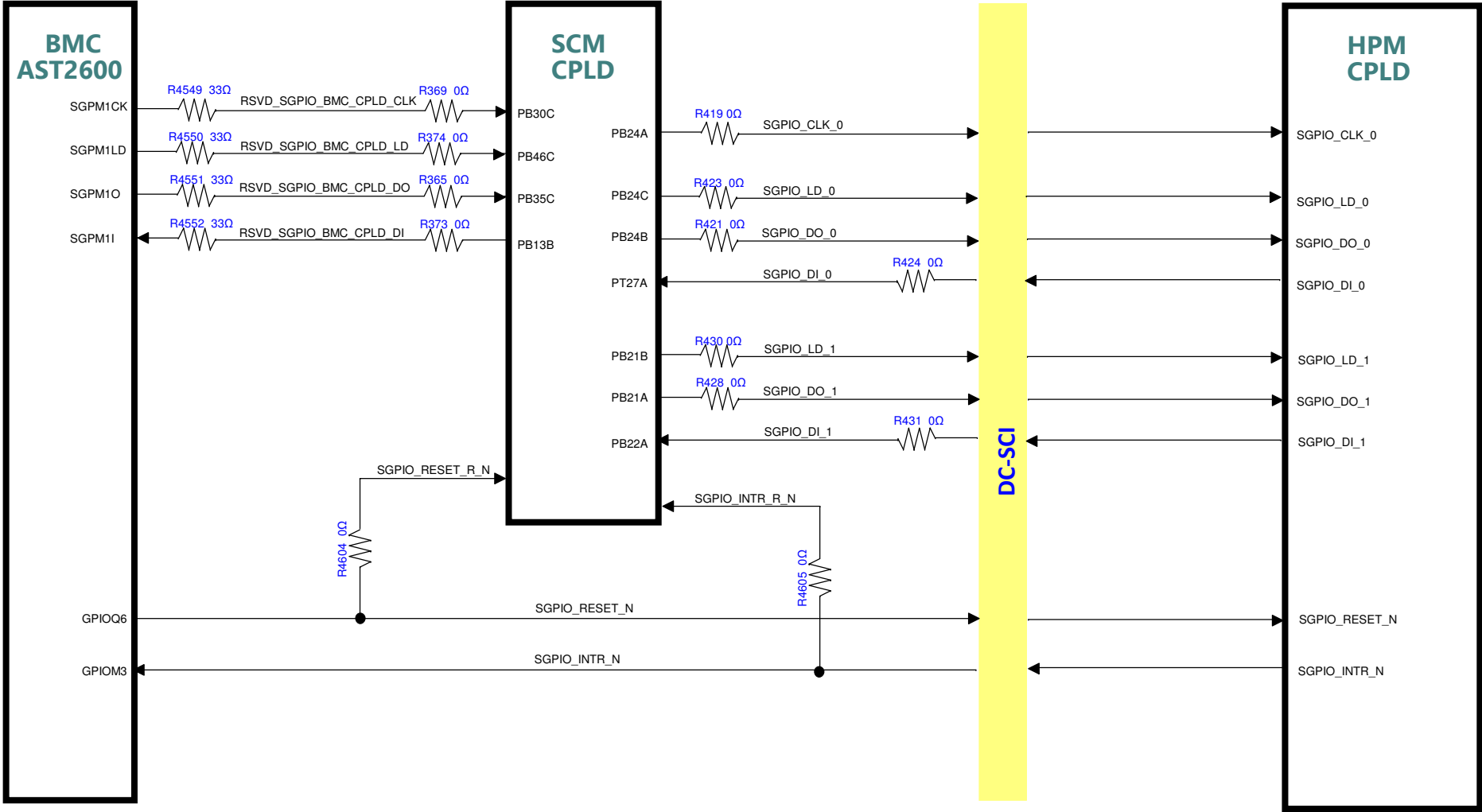
CLOCK_TOPOLOGY (AMD)



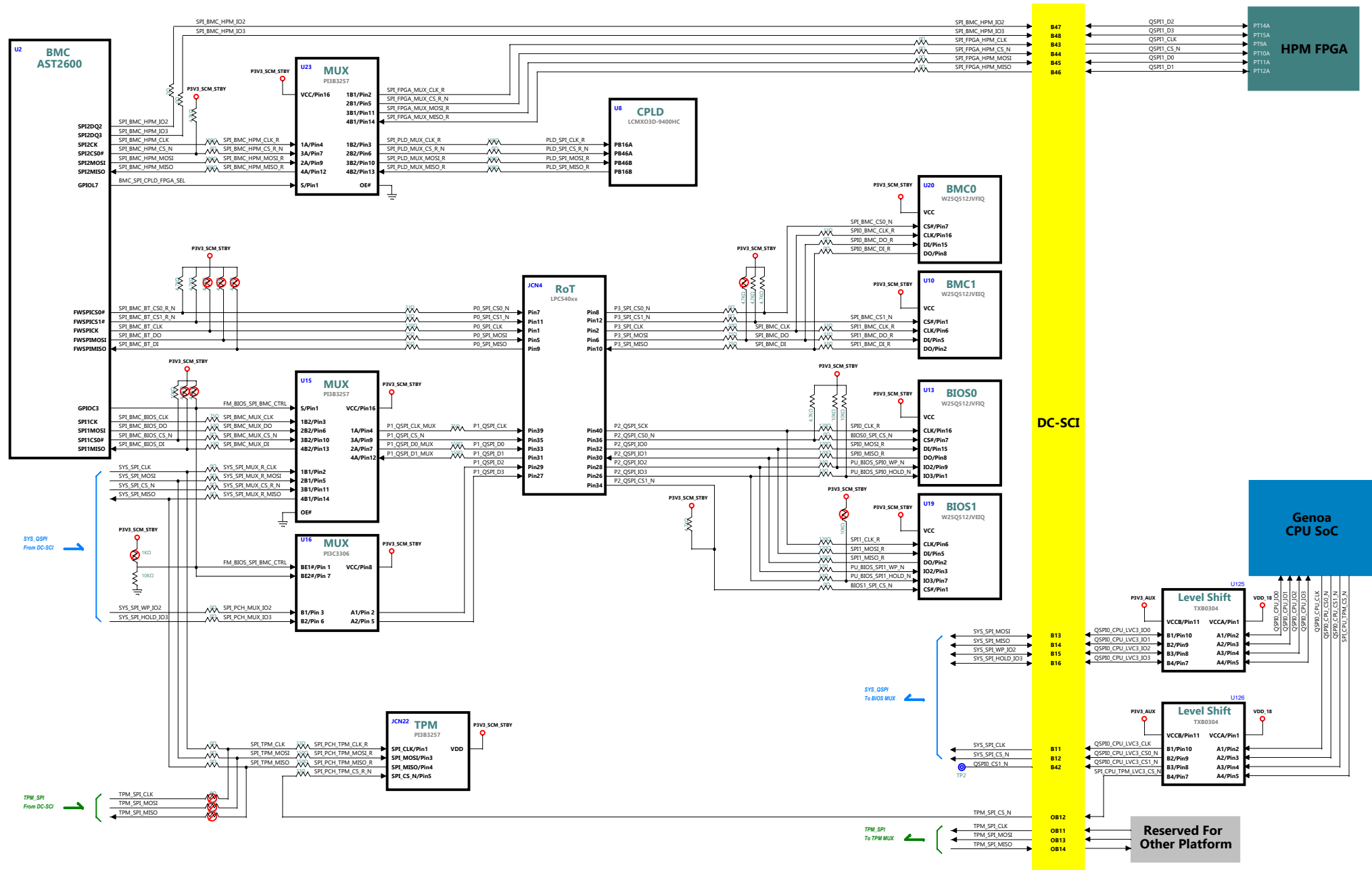
I2C_BLOCK_DIAGRAM - BMC/SOC (AMD)



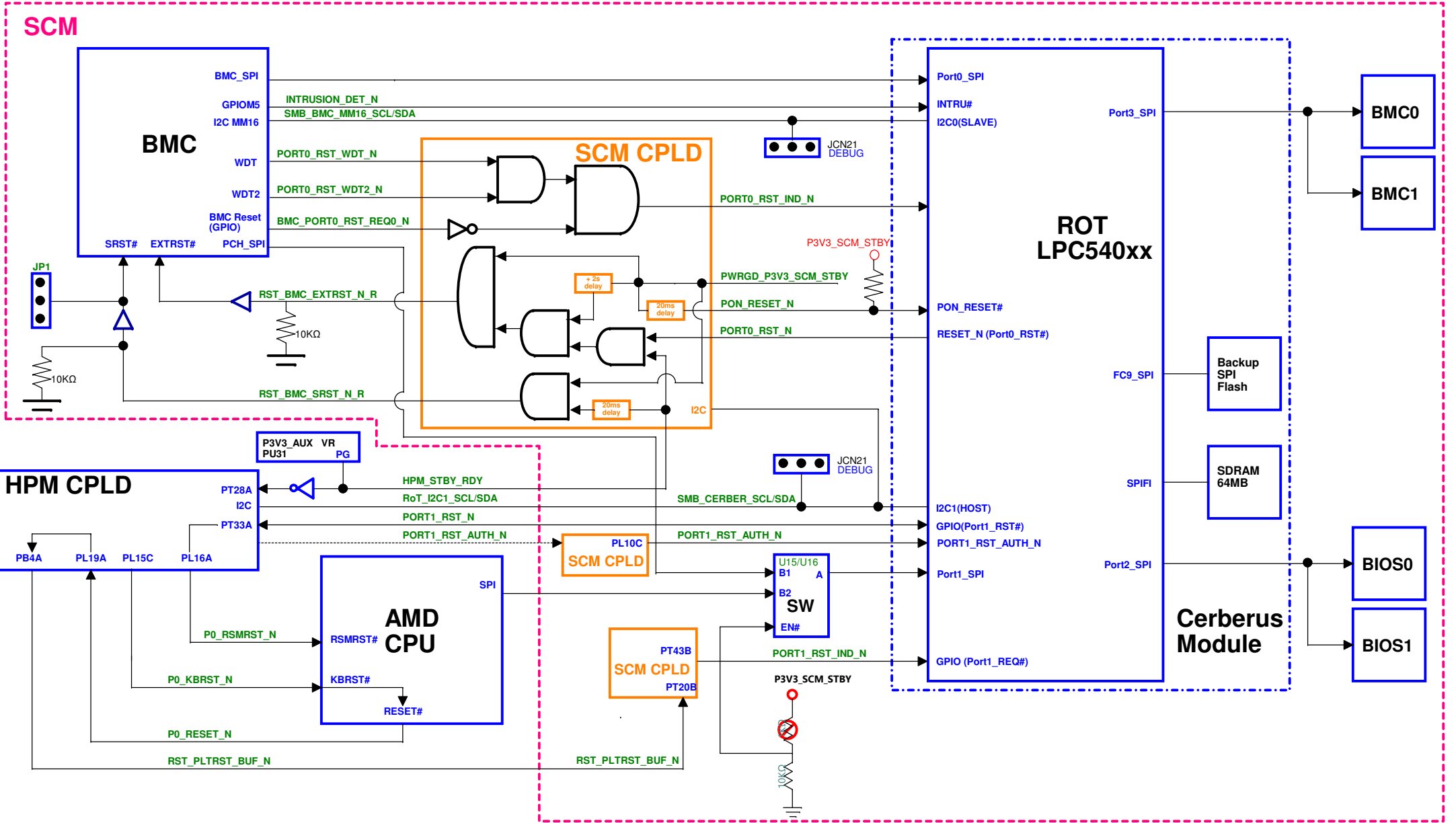
SGPIO_BLOCK_DIAGRAM



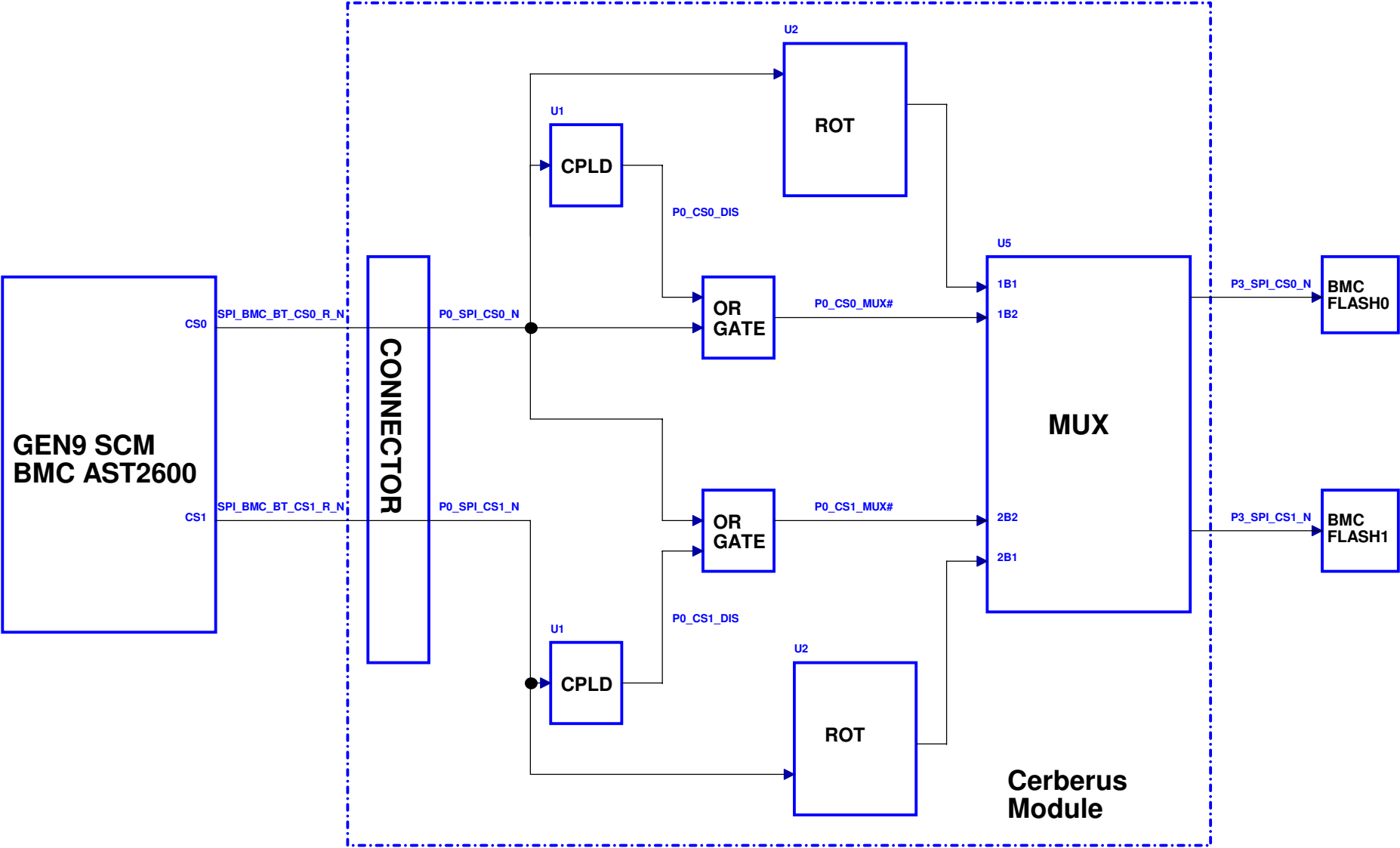
SPI BLOCK DIAGRAM (AMD)



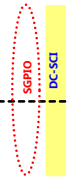
CERBERUS_BLOCK_DIAGRAM (AMD)



CERBERUS CS BLOCK DIAGRAM_BMC



--- Via CPLD SGPIO



--- Via CPLD SGPIO

RM THROTTLE#
Pin. 5

JCN57

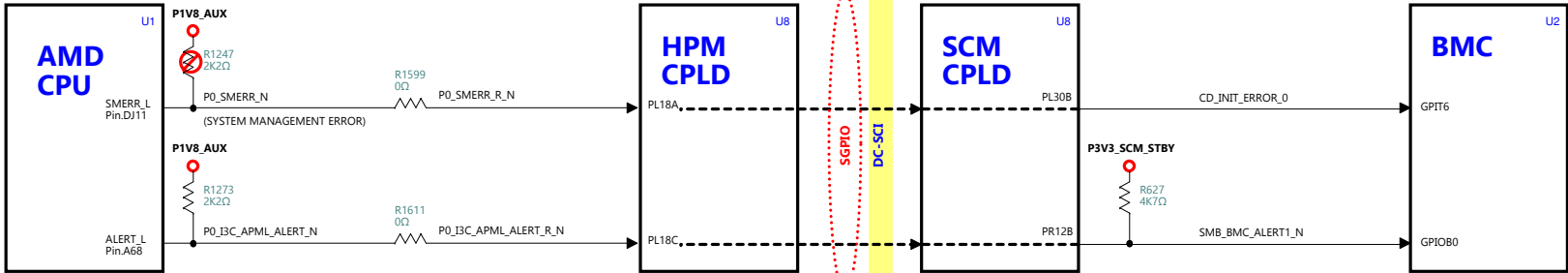
PSU2 ALERT#
Pin. 11

JCN58

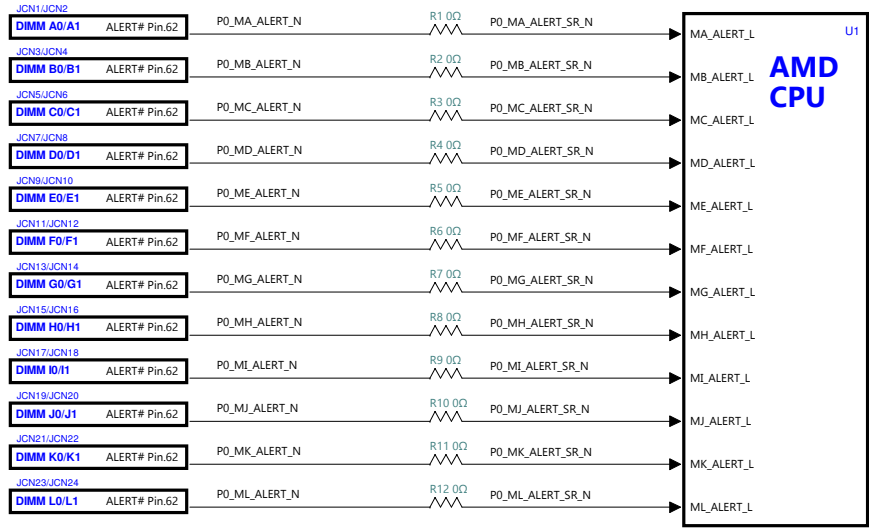


SYSTEM_ERROR_REPORTING (AMD)

----- Via CPLD SGPIO

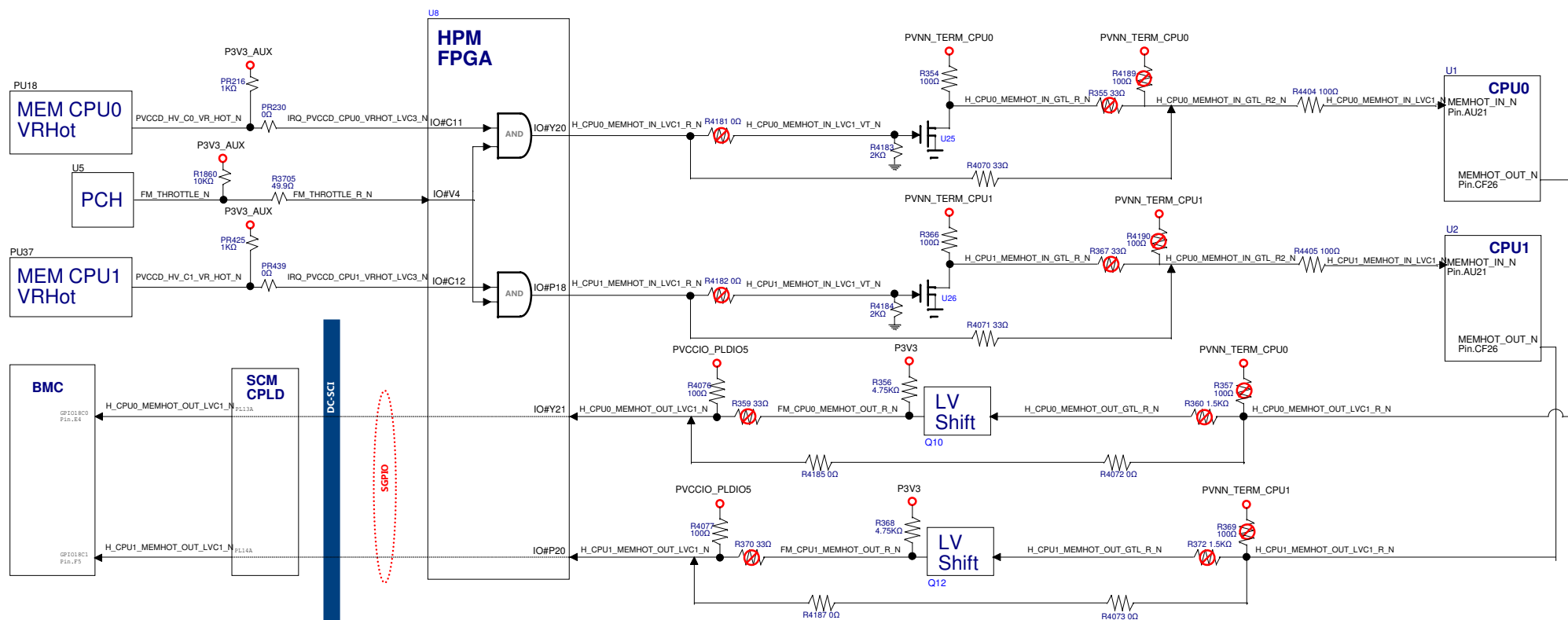


DIMM ALERT FOR CRC ERROR (CYCLIC REDUNDANCY CEHCK)



Reserved for INTEL

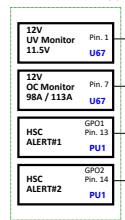
..... Via CPLD SGPIO



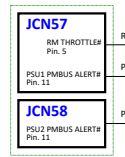
PROCHOT DIAGRAM (AMD)

..... Via CPLD sGPIO

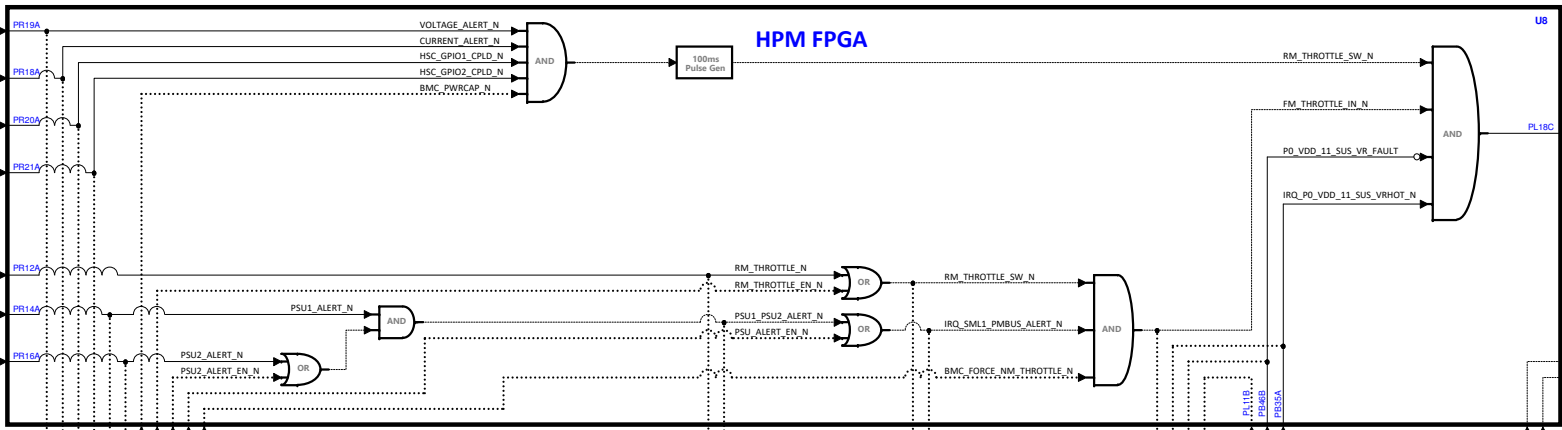
FAST PROCHOT Triggers



Power Cap Triggers

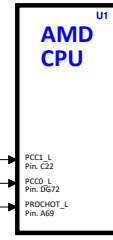
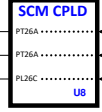
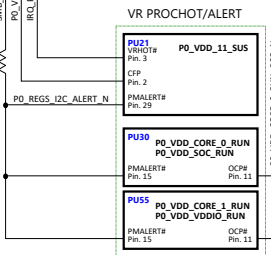
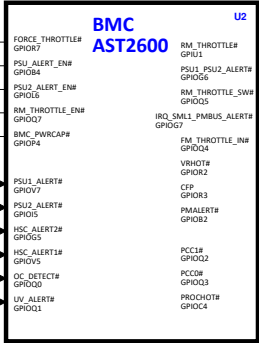
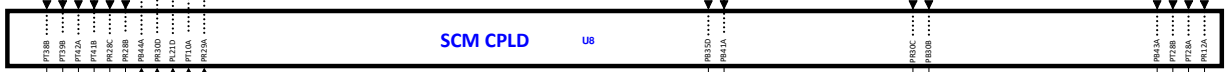


sGPIO

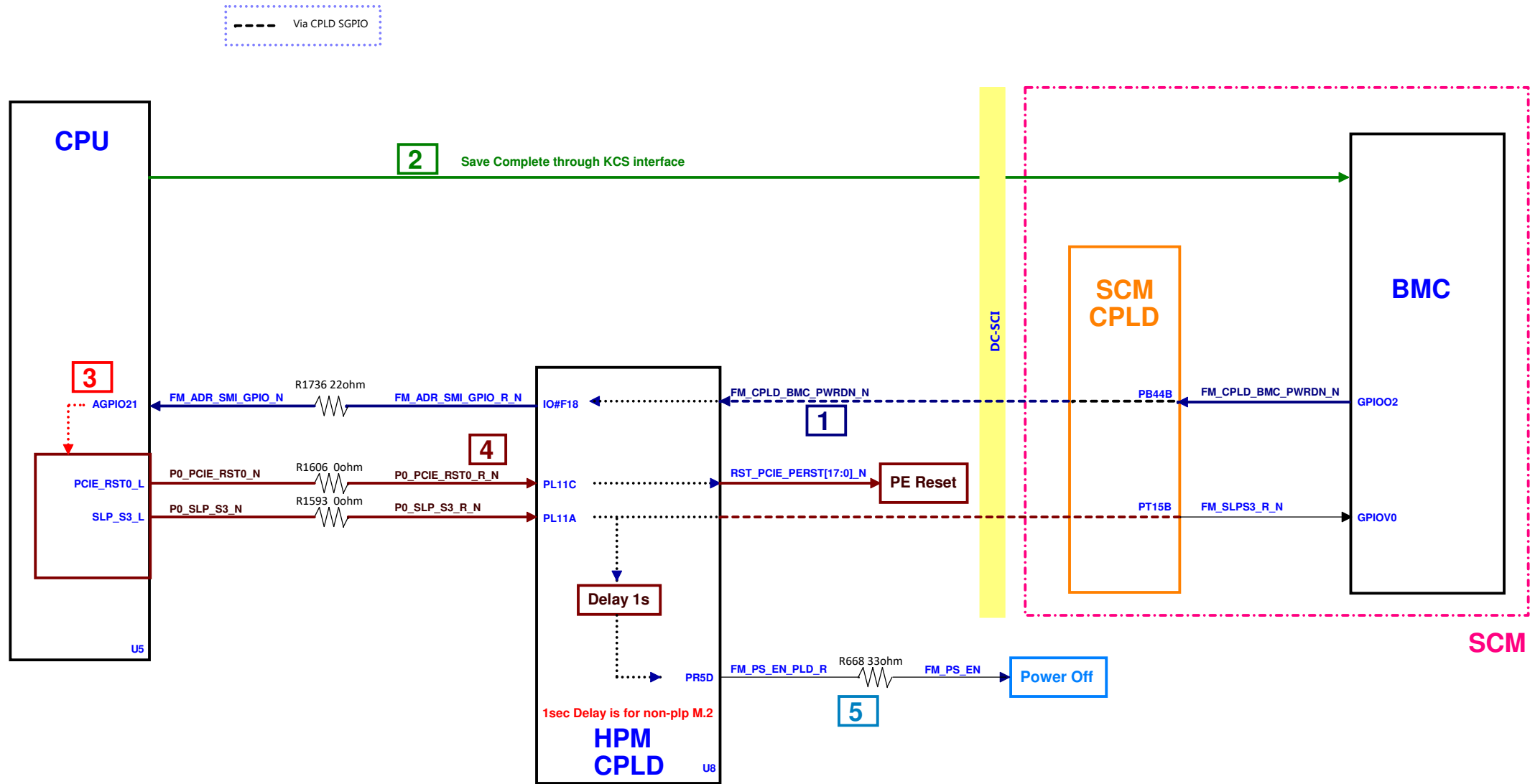


HPM
SCM

sGPIO



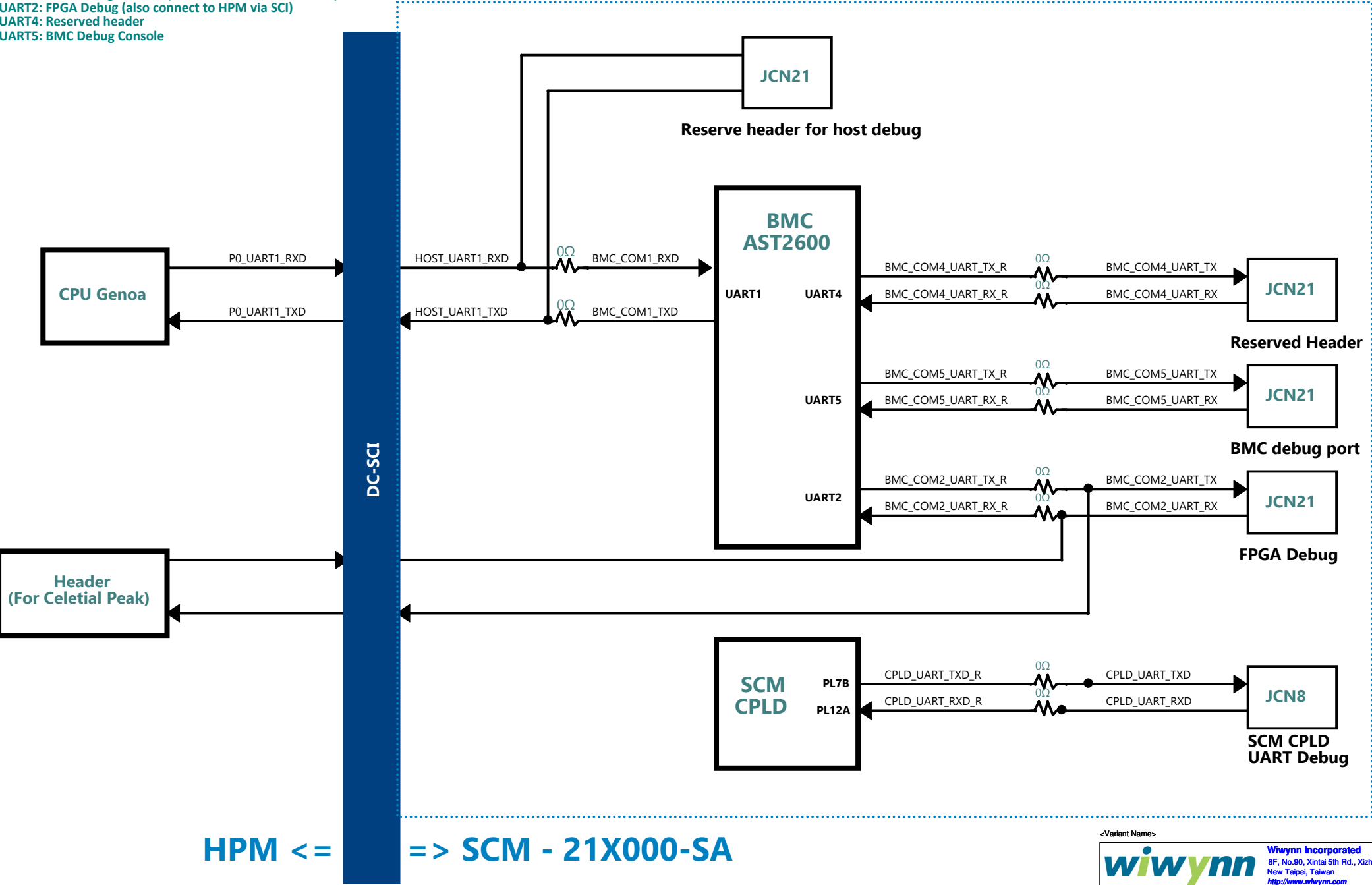
SW_NVDIMM_BLOCK_DIAGRAM Reserved for INTEL



1. FM_CPLD_BMC_PWRDN_N to CPLD which triggers SW_NVDIMM to CPU
2. Save Complete through KCS interface
3. CPU ADR SMI
4. RESET# and SLP_S3# asserted

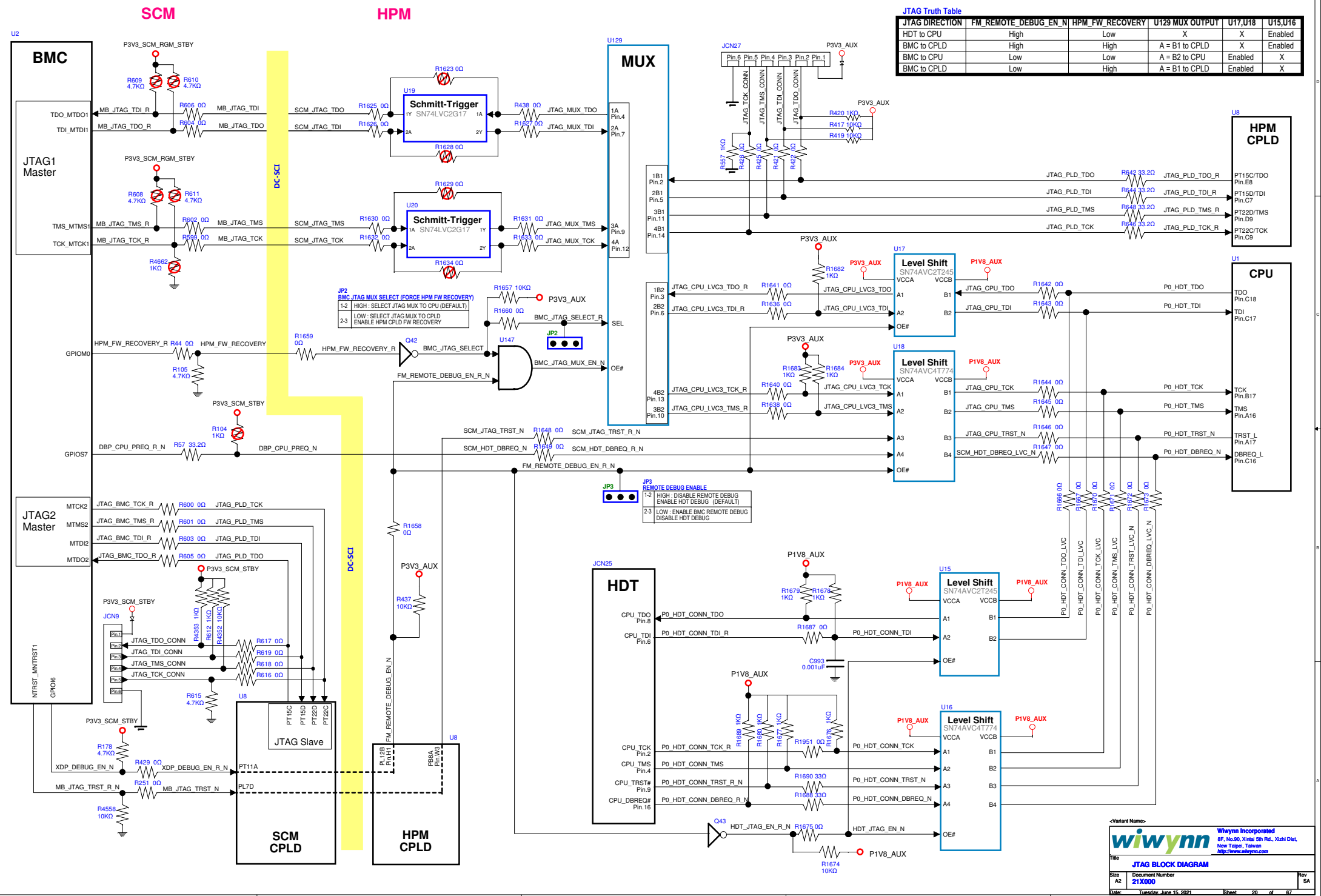
UART BLOCK DIAGRAM (AMD)

UART1: Host Debug Console (also connect to HPM via SCI)
UART2: FPGA Debug (also connect to HPM via SCI)
UART4: Reserved header
UART5: BMC Debug Console



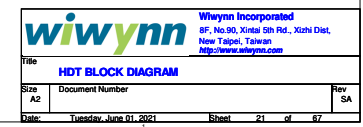
HPM <= => SCM - 21X000-SA

JTAG BLOCK DIAGRAM (AMD)



4

----- Via CPLD SGP10



STACKUP

Board Number:		21X000-SA	Version		03
Project Name:		C2109			
Model Name:		POC SCM			
Layer Count:		8			
Date:		2021/7/2			
Middle Loss Material:		IT-1706RA1/NPG-171/EM-8285+RTF (Typ) 70, Td=305)			
Gold Finger (V/N):		Yes (see Note 10)			
Thickness (mm):		1.57			
Product Type:		Halogen Free(HF)			
E/F / S Engineer:		Dylan / Hank			
Black Drill Via / Panel Rotation 10°		No / No			

Layer	Cu Weight	Thickness	Glass/Copper Type	Er (1GHz)	DF @4GHz 80C
Wiwynn/Vendor		Wiwynn/Vendor	Wiwynn/Vendor	Wiwynn/Vendor	Wiwynn/Vendor
Mask		0.5			0.035282
Signal	0.5 oz + plating	1.9	HTE	3.5	
Prepreg		2.7			0.02842
Ground	1.0 oz	1.2	RTF	3.5	
Core		3		3.5	0.019
Signal	1.0 oz	1.2	RTF	3.5	
Prepreg		17.8			0.019
Ground	1.0 oz	1.2	B IT	3.5	0.019
Core		3		3.5	0.019
Signal/Power	1.0 oz	1.90	RTF		
Prepreg		17.80	0	3.5	0
Signal	1.0 oz	1.20	RTF		
Core		3.00	0	3.5	0
Signal	1.0 oz	1.20	B IT		
Prepreg		2.20	0	3.5	0
Signal	0.5 oz + plating	1.90	HTE		
Mask		0.50	0	3.8	0
Thickness requirement : 1.57 ± 10% (mm)		mm (without solder mask)	65.00	0.00	
		mm (without solder mask)	1.95	0.00	

Single Ended Impedance (ohm)											
S0						S5					
Inner/Outer+/ 10%						Inner/Outer+/ 10%					
Misc V/D, SE clocks						BMC DDR					
SE						SE					
1, 3, 5, 6, 8						1, 3, 5, 6, 8					
Wiwynn		Vendor		Wiwynn		Vendor		Wiwynn		Vendor	
Width	Imp	Width	Imp	Width	Imp	Width	Imp	Width	Imp	Width	Imp
4.9(L2)	56	?	?	5.50(3)	54.6	?	?	?	?	?	?
reference layer						reference layer					
4.20(3,4,5)	49.05	?	?	3.50(20,4)	55.36	?	?	?	?	?	?
reference layer						reference layer					
4.20(4,7)	50.37			3.50(40,7)	54.36						
4.20(5,7)	49.05	?	?	3.50(50,7)	55.36	?	?	?	?	?	?
reference layer						reference layer					
4.3(L1)	56	?	?	3.30(7)	54.6			?	?	?	?

Differential Impedance (ohm)											
B5						300					
Inner/Outer+/ 10%						Inner/Outer+/ 10%					
Diff clocks, HS Diff, PCIe, UPI, DMI, USB, SATA						KR 1GBase					
DP						DP					
1, 3, 5, 6, 8						1, 3, 6, 8					
Wiwynn		Vendor		Wiwynn		Vendor		Wiwynn		Vendor	
Width	Space	Imp	Width	Space	Imp	Width	Space	Imp	Width	Space	Imp
4(L2)	4	85	?	?	?	4(L3)	6	100	?	?	?
reference layer						reference layer					
4(L3,4,5)	4	85	?	?	?	3.50(3,4)	7	100	?	?	?
reference layer						reference layer					
4(L4,7)	4					3.50(4,6,7)	7				
4(L5,7)	4	85	?	?	?	3.50(5,6,7)	7	100	?	?	?
reference layer						reference layer					
4(L1)	4	85	?	?	?	4(L7)	6	100	?	?	?

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Title
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Size A3
Document Number
21X000

Rev
SA

Date: Friday, May 28, 2021

Sheet 22 of 67

SCM_JUMPER_SETTING/LED_DEFINE/Header

LED

Location	Color	Function
LED4	Red	CATERR LED
LED5	Green	BMC Heartbeat
LED7	Green/Amber	Power status
LED8	Blue	UID
LED9	Red	Blade Attention
LED10	Green	CPLD Heartbeat

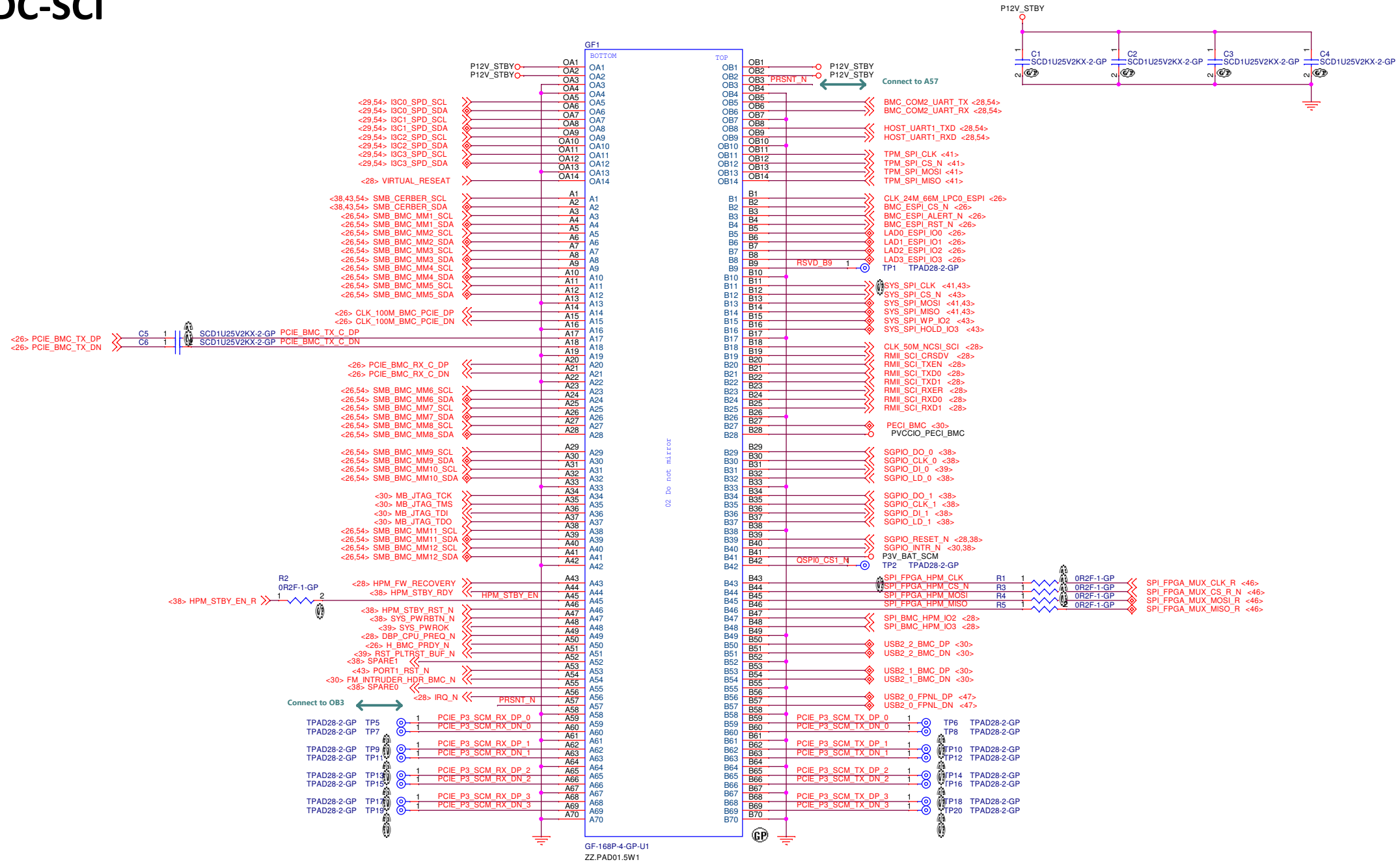
Debug Header

Location	Function
JCN20	I2C[9:1] Debug Header
JCN21	I2C[16:15] / I2C[10] / RoT I2C / UART[2:1] / UART[5:4] Debug Header
JCN23	I3C[4:1] Debug Header

Jumper

Location	Status	Function
JP1	1-2 Normal (Default)	When jumper is set to 2-3, SRST# on BMC is pulled low and all pins on BMC are set to high impedance to disable BMC.
	2-3 Hold BCM in Reset	
JP2	1-2 eMMC RST Disable (Default)	
	2-3 eMMC RST Enable	

DC-SCI



DESIGN NOTE:

1. QSPI0_CS1_N un-used with Cerberus, so connect to TP.
2.x4 PCIe RFU.

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Title			
DC-SCI			
Size	Document Number	Rev	
A3	21X000	SA	
Date:	Friday, May 28, 2021	Sheet	24 of 67

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<Variant Name>



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Title

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Size

A3

Document Number

21X000

Rev

SA

Date:

Friday, May 28, 2021

Sheet

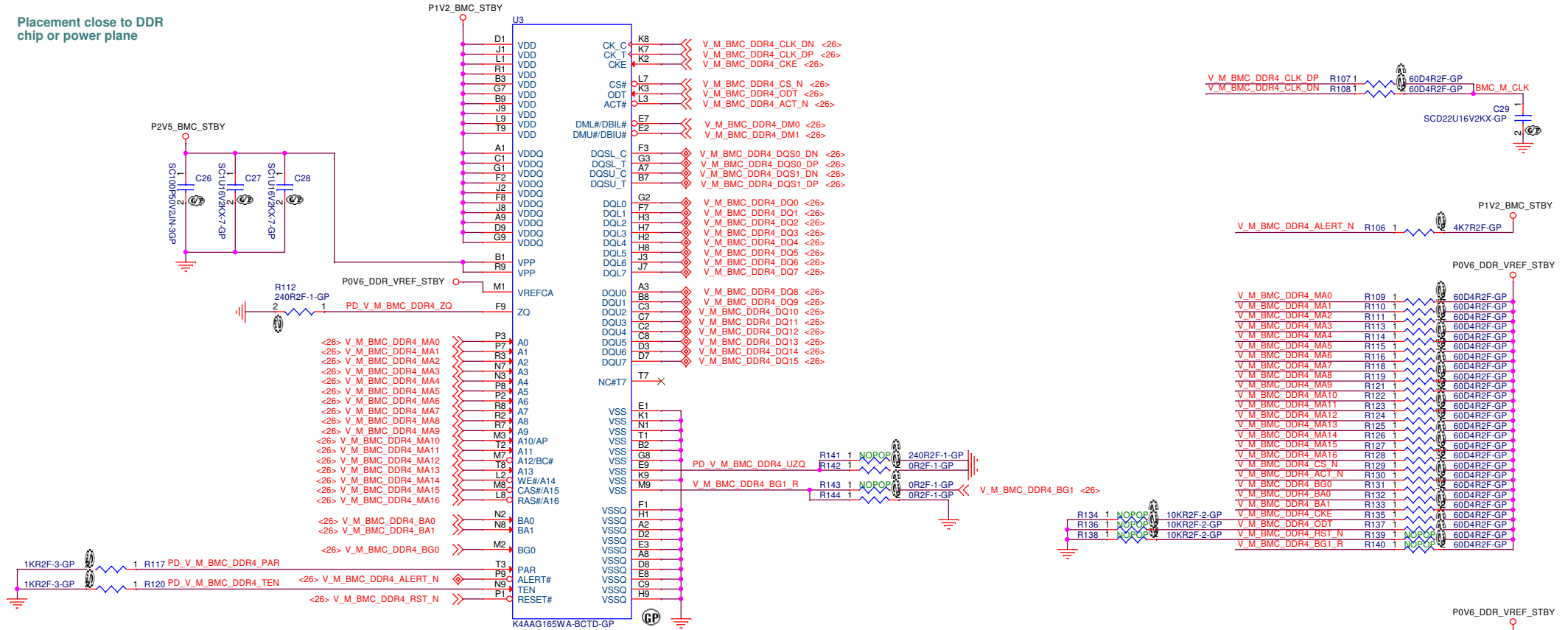
25

of

67

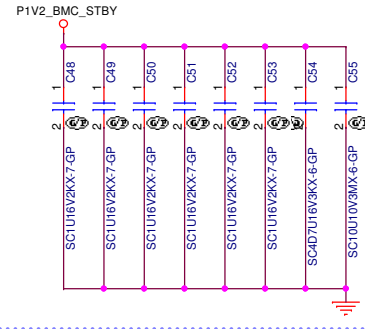
BMC_DDR

Placement close to DDR
chip or power plane



POWER DE-COUPLE CAP

Placement close to DDRVDDQ Termination



SINGLE/TWIN DIE SELECTION

	Single Die Part	Twin Die Part
R141/R142	R142 0 ohm POP	R141 240 ohm (1%) POP
R143	NOPOP	POP
R144	POP	NOPOP
R140	NOPOP	POP

<Variant Name>

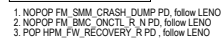
wiwynn Wiwynn Incorporated
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Title **BMC_DDR**

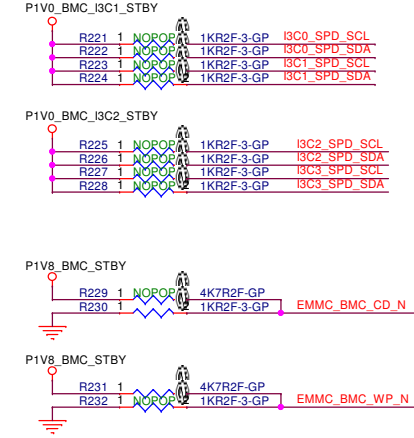
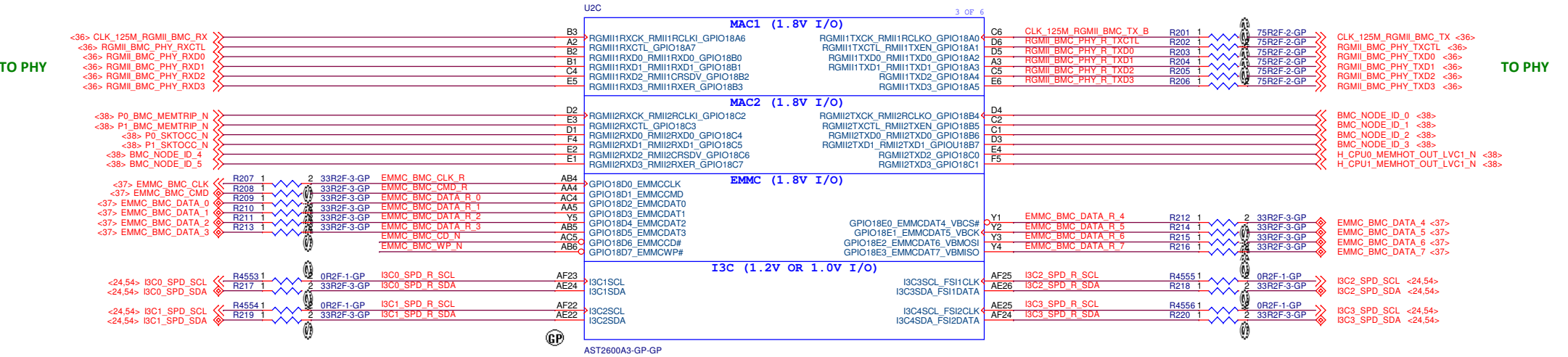
Size A3 Document Number **21X000** Rev SA

Date: Friday, May 28, 2021 Sheet 27 of 67

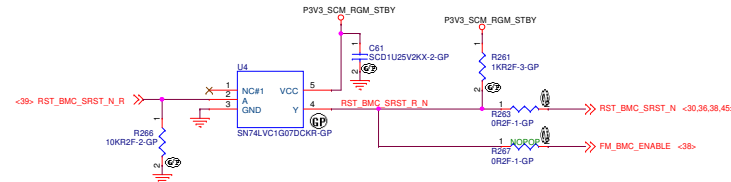
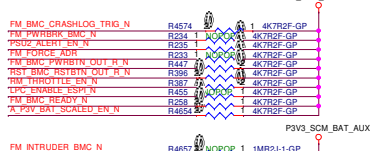
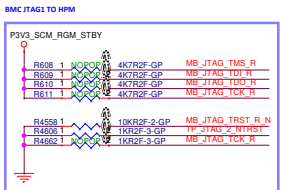
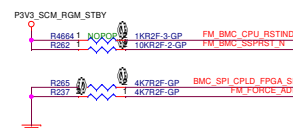
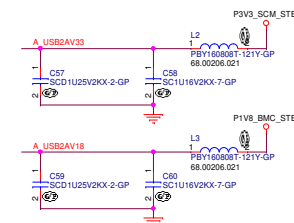
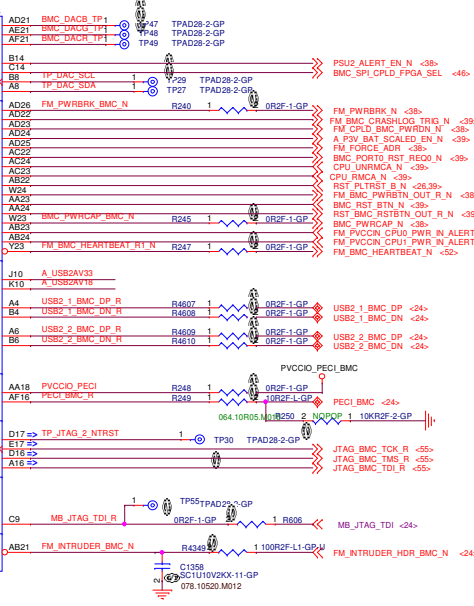
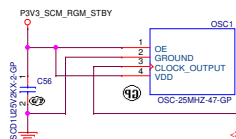
To NCSI on MB



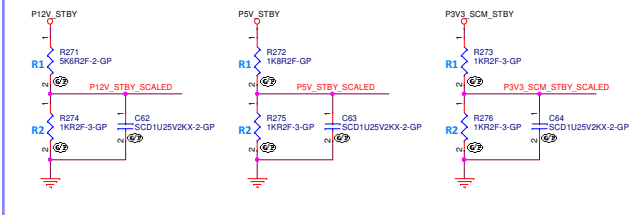
BMC_MAC



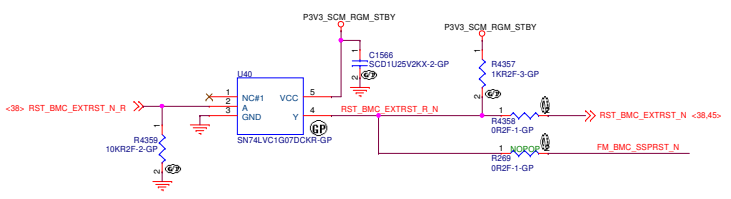
BMC_ADC



ADC input voltage range : 0 ~ 2.5V
A good nominal input voltage = 3/4 V_{I_max} = 1.8V
 $(ADC_Value/1023)*2.5 = V^- + (V^+ - V^-)*R2 / (R1+R2)$

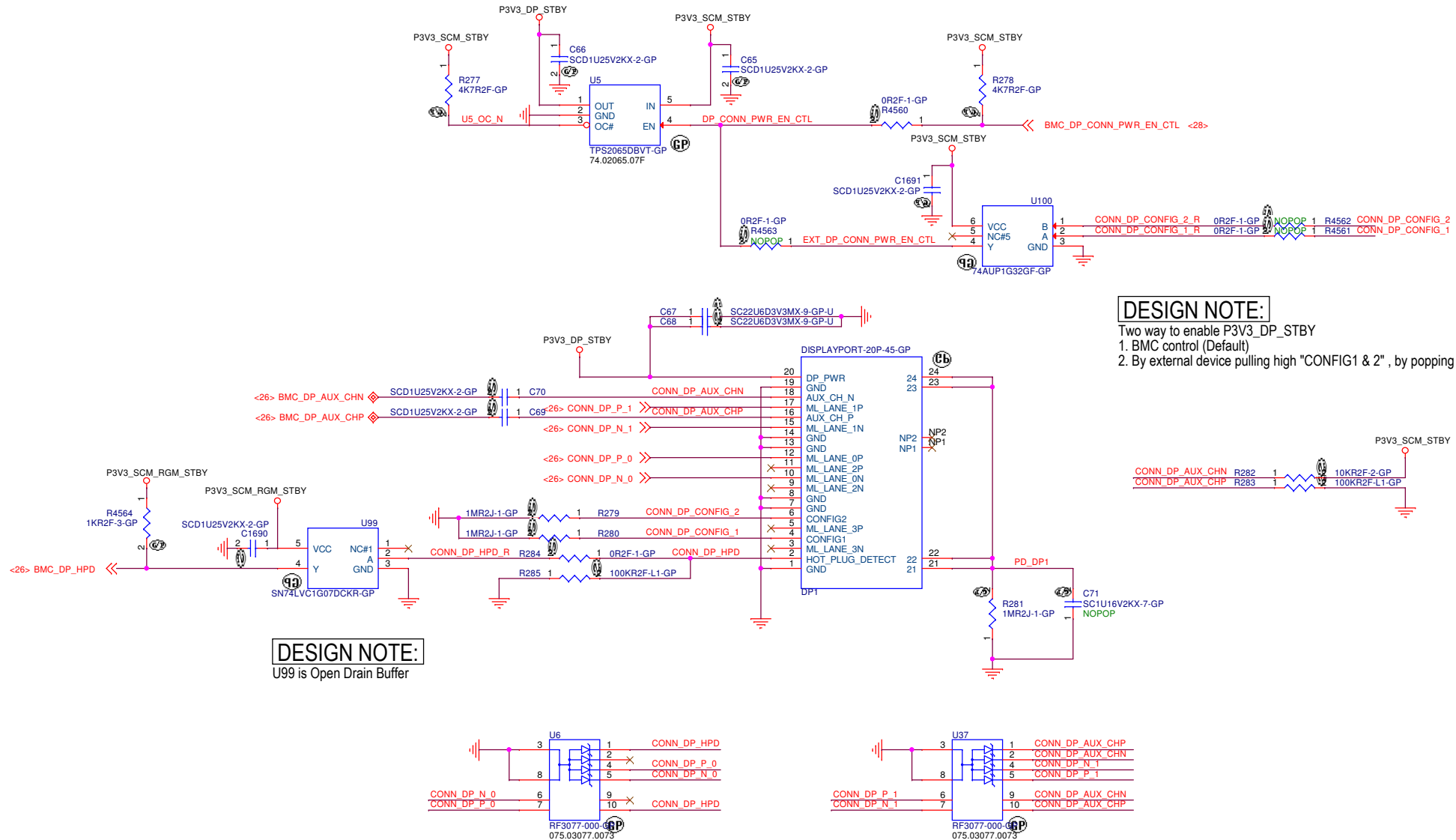


DESIGN NOTE:

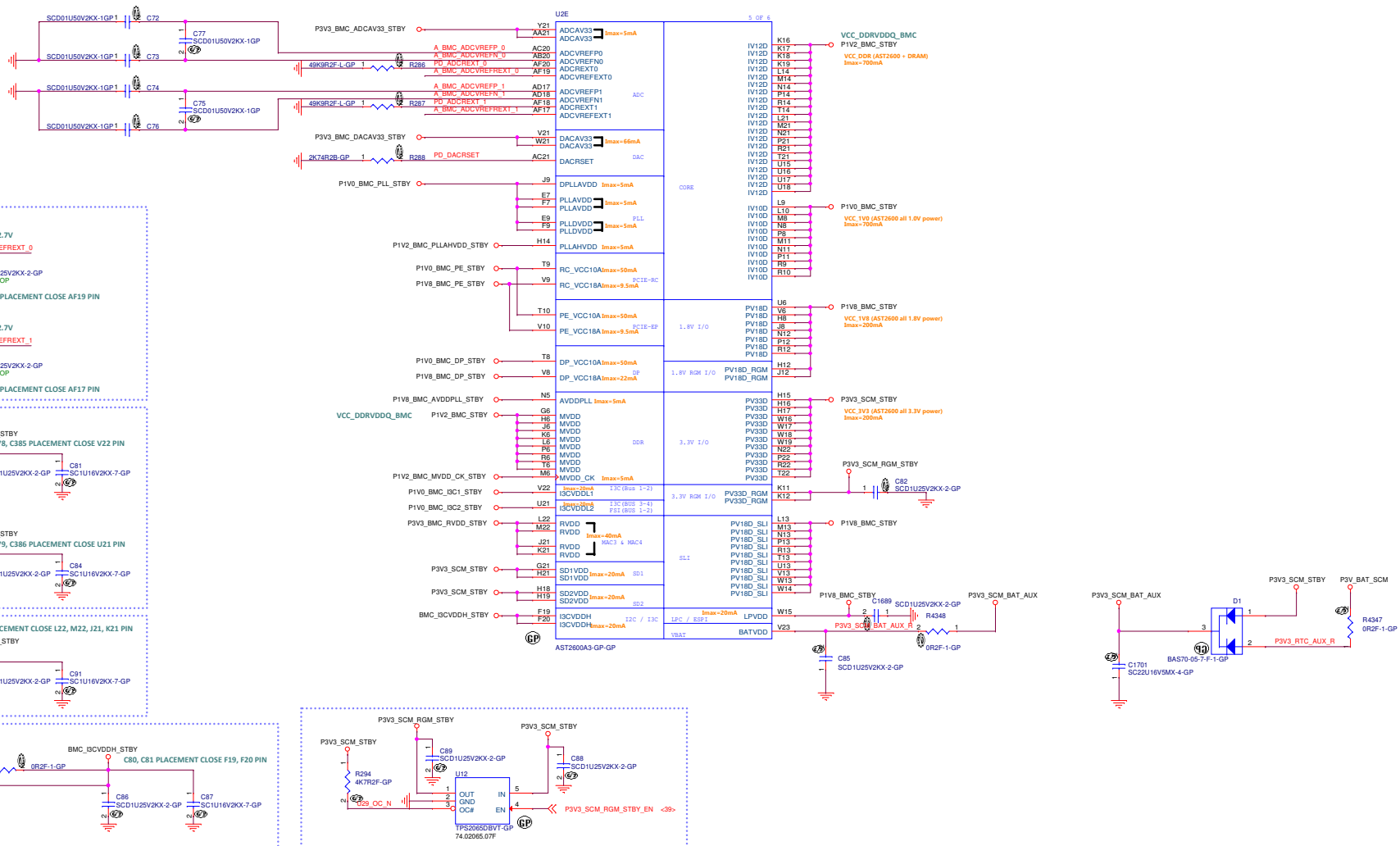


<Variant Name>

BMC_DISPLAY_PORT (5/10)



BMC_POWER



DESIGN NOTE:

1. POP R4666, NOPOP R4667 for 3.3V (default)
2. NOPOP R4666, POP R4667 for 1.8V

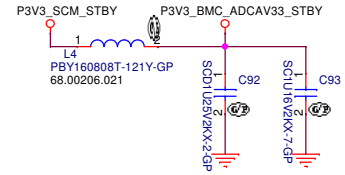
<Variant Name>

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New Taipei, Taiwan
<http://www.wiwynn.com>

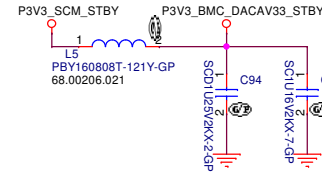
Title			Rev
BMC_POWER			SA
Size	Document Number		
A2	21X000		
Date:	Friday, May 28, 2021	Sheet	32 of 57

BMC_POWER_CAP

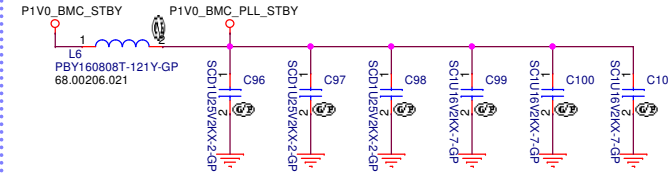
ADC ANALOG POWER PLACEMENT CLOSE Y21, AA21 PIN OR POWER PLANE



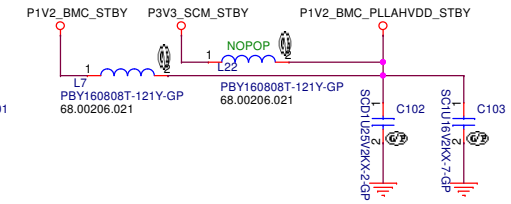
DAC ANALOG POWER PLACEMENT CLOSE V21, W21 PIN OR POWER PLANE



BMC PLL ANALOG POWER PLACEMENT CLOSE J9, E7, F7, E9, F9 PIN OR POWER PLANE

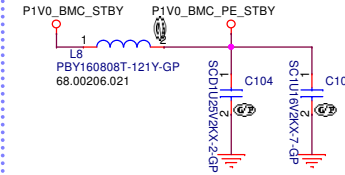


PLACEMENT CLOSE H14 PIN OR POWER PLANE

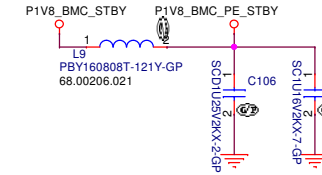


REMOVE PCIE-RC ANALOG POWER

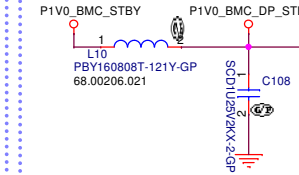
PCIE-PE ANALOG POWER PLACEMENT CLOSE T10 PIN OR POWER PLANE



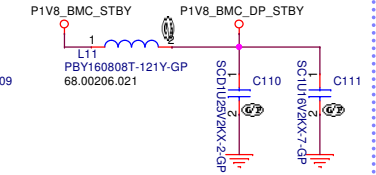
PLACEMENT CLOSE V10 PIN OR POWER PLANE



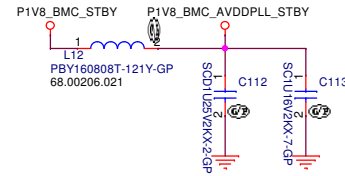
DP ANALOG POWER PLACEMENT CLOSE T8 PIN OR POWER PLANE



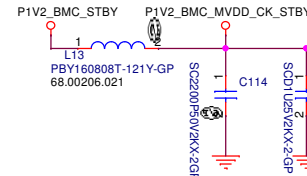
PLACEMENT CLOSE V8 PIN OR POWER PLANE



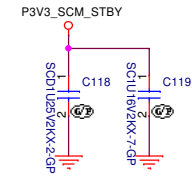
DDR PHY PLL & CK POWER PLACEMENT CLOSE N5 PIN OR POWER PLANE



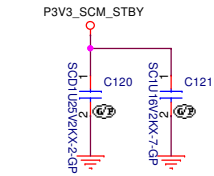
PLACEMENT CLOSE M6 PIN OR POWER PLANE



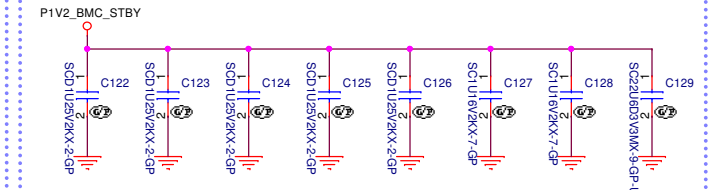
SD I/O POWER PLACEMENT CLOSE G21, H21 PIN OR POWER PLANE



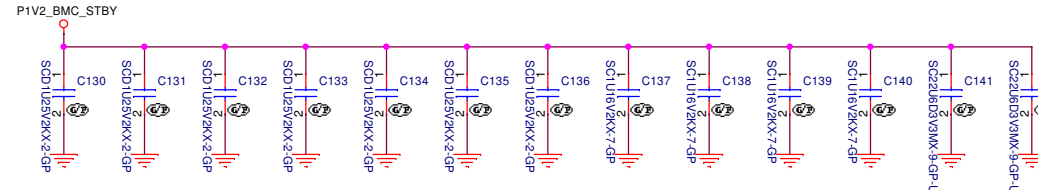
PLACEMENT CLOSE H18, H19 PIN OR POWER PLANE



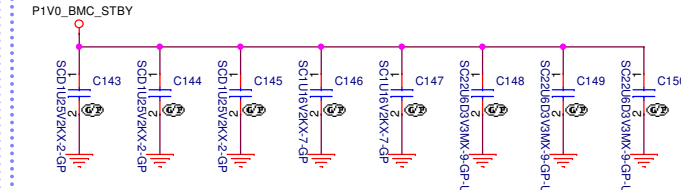
PLACEMENT CLOSE G6, H6, J6, K6, L6, P6, R6, T6 PIN OR POWER PLANE



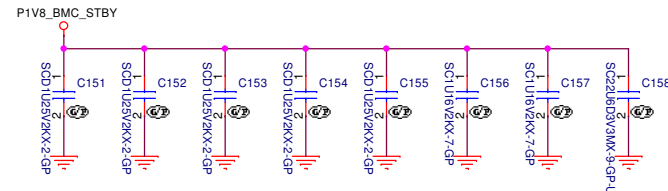
PLACEMENT CLOSE K16, K17, K18, K19, L14, M14, N14, P14, R14, T14, L21, M21, N21, P21, R21, T21, U15, U16, U17, U18 PIN OR POWER PLANE



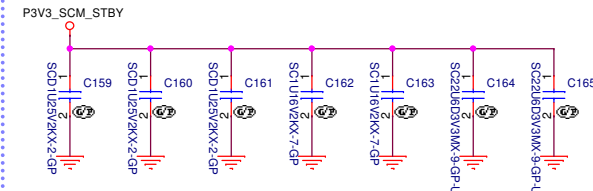
PLACEMENT CLOSE L9, L10, M8, N8, P8, M11, N11, P11, R9, R10 PIN OR POWER PLANE



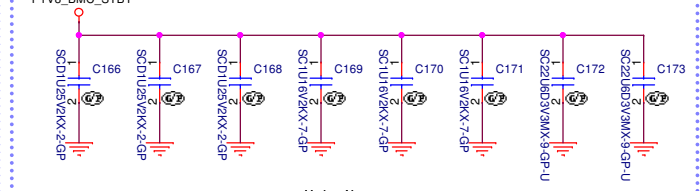
PLACEMENT CLOSE U6, V6, H8, J8, N12, P12, R12, H12, J12 PIN OR POWER PLANE



PLACEMENT CLOSE H15, H16, H17, W16, W17, W18, W19, N22, P22, R22, T22 PIN OR POWER PLANE



PLACEMENT CLOSE L13, M13, N13, P13, R13, T13, U13, V13, W13, W14 PIN OR POWER PLANE



<Variant Name>

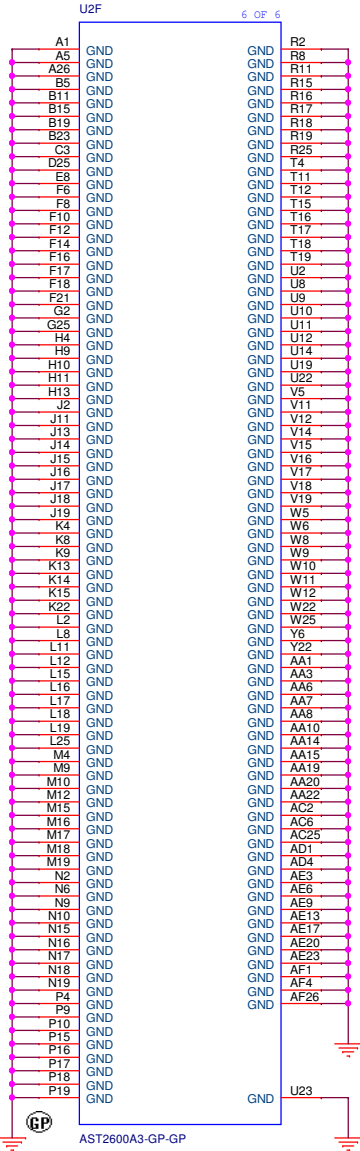
wiwynn Wiwynn Incorporated
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New Taipei, Taiwan
<http://www.wiwynn.com>

Title **BMC_POWER_CAP**

Size A3 Document Number **21X000** Rev SA

Date: Friday, May 28, 2021 Sheet 33 of 67

BMC_GND



BMC_STRAPS (9/10)

DESIGN NOTE:

AST2600 HARDWARE STRAP OPTION TABLE:

OPTION1: 11 dedicated strap (SCU510[30]=1)

OPTION2: 5 dedicated strap (SCU510[30]=0)

TXD5: embedded internal pull high resistor of value 26K~71K.

FWSPICK, FWSPIMOSI, FWSPIMISO: embedded internal pull low of value 33K~89K.

GPIOY[5:4], GPIOZ[7:3]: embedded internal pull low of value 33K~89K.

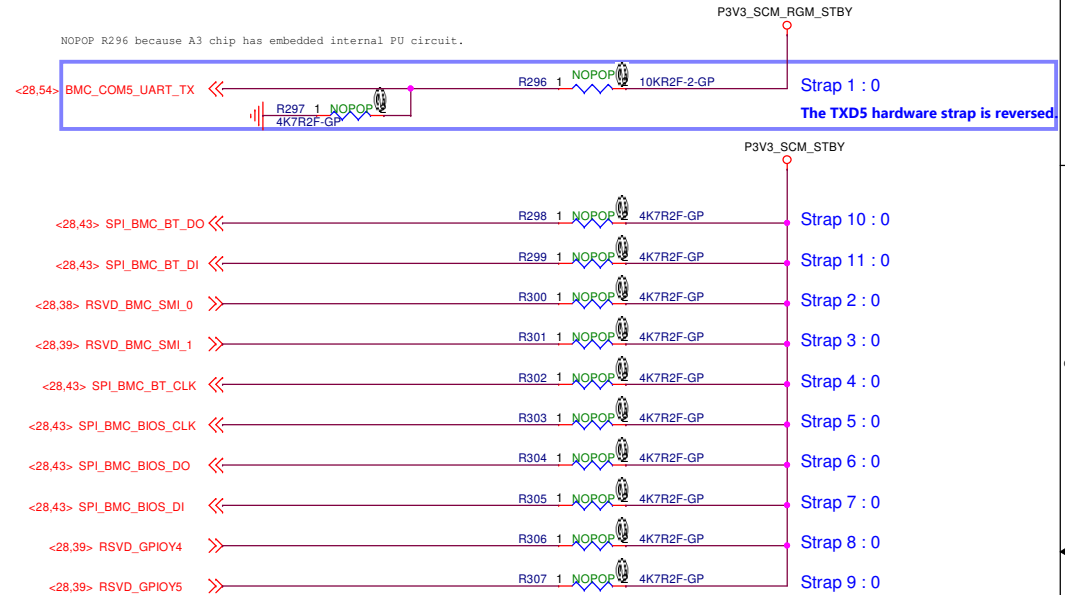
11 dedicated strap function pins when SCU510[30]= 1. Listed as below:

STRAP NUM	BALL NUM	BALL NAME	VAL	DESCRIPTION
1	C8	TXD5	--	DISABLE CPU BOOT 0: ENABLE BOOT (DFLT) (PU) 1: DISABLE CPU OPERATION, WHEN NO FIRMWARE EXIST
2	AD11	GPIOZ6	--	BOOT SPI FLASH SIZE 000: NO DEFINE SIZE 100: 16MB
3	AF10	GPIOZ7	--	001: 2MB 101: 32MB
4	AF13	FWSPICK	--	010: 4MB 110: 64MB
			--	011: 8MB 111: 128MB
5	AB11	GPIOZ3	--	HOST SPI FLASH SIZE 000: NO DEFINE SIZE 100: 16MB
6	AC11	GPIOZ4	--	001: 2MB 101: 32MB
7	AA11	GPIOZ5	--	010: 4MB 110: 64MB
			--	011: 8MB 111: 128MB
8	AE12	GPIOY4	--	HOST SPI CRIM 00: DISABLE CRIM 01: 1MB 10: 2MB 11: 4MB
9	AF12	GPIOY5	--	
10	AC14	FWSPIMOSI	--	ENABLE SECURE BOOT BY PIN STRAP 0: DISABLE 1: ENABLE
11	AB13	FWSPIMISO	--	ENABLE LOW SECURE BOOT FUNCTION 0: DISABLE 1: ENABLE

5 dedicated strap function pins when SCU510[30]= 0 Listed as below:

STRAP NUM	BALL NUM	BALL NAME	VAL	DESCRIPTION
1	C8	TXD5	0	DISABLE CPU BOOT 0: ENABLE BOOT (DFLT) (PU) 1: DISABLE CPU OPERATION, WHEN NO FIRMWARE EXIST
5	AB11	GPIOZ3	0	Disable VGA device (PCIe device 0) 0: Enable VGA device 1: Disable VGA device
6	AC11	GPIOZ4	0	Select LPC/eSPI 0: eSPI mode (DFLT) 1: LPC mode
7	AA11	GPIOZ5	0	Disable HBLed 0: Enable Heartbeat LED 1: Disable Heartbeat LED
2	AD11	GPIOZ6	0	Enable GPIO Pass Through 0: Disable, pass through. (default) 1: Enable pass-through at power on. GPIO0 -> GPIO1, GPIO2 -> GPIO3, GPIO4 -> GPIO5
3	AF10	GPIOZ7	0	Enable ACPI function 0: Disable ACPI (default) 1: Enable ACPI
4	AF13	FWSPICK	0	Enable Boot from Uart5 by Pin Strap 0: Disable (DFLT) 1: Enable boot from Uart5
10	AC14	FWSPIMOSI	0	ENABLE SECURE BOOT BY PIN STRAP 0: DISABLE (DFLT) 1: ENABLE
11	AB13	FWSPIMISO	0	ENABLE LOW SECURE BOOT FUNCTION 0: DISABLE (DFLT) 1: ENABLE

7/27 Correct the value of strap 4 & 6 to 0.

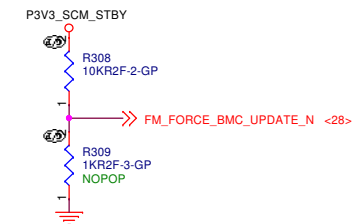


DESIGN NOTE:

BMC FORCE UPDATE

When no firmware existed,
BMC must be disabled to
prevent mal-function.

High	NORMAL OPERATION (DEFAULT)
Low	FORCE BMC UPDATE



PD 50K ohm on GPIOZ4. Base on
Azure_SCM_SPEC_V1.6, only support
ESPI.

DESIGN NOTE:

LPC/ESPI SELECT

BOM select LPC/ESPI

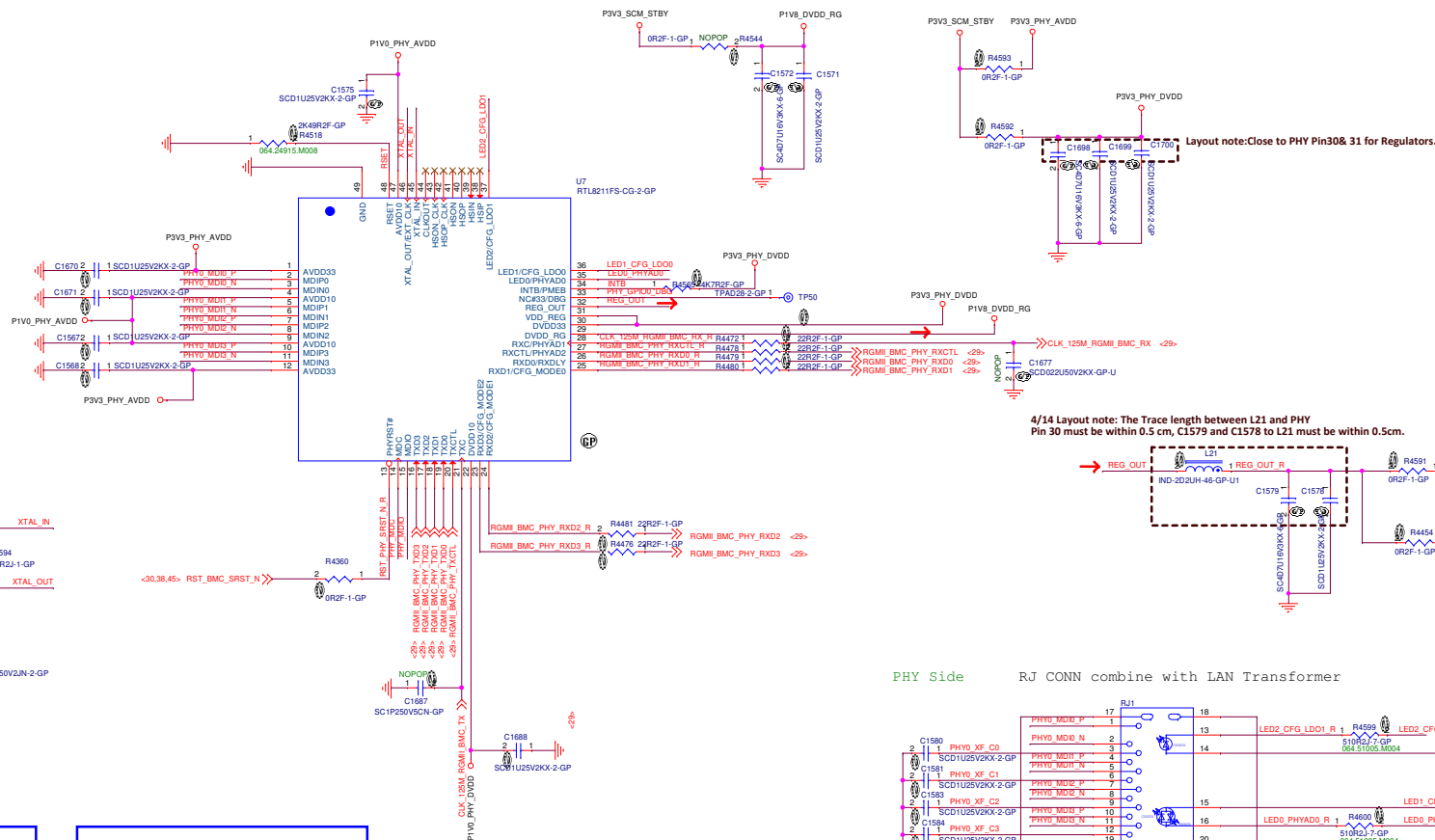
POP R302/NOPOP R4356	LPC
NOPOP R302/POP R4356	ESPI

3/30 Update Dedicated strap function pins table when
SCU510[30]=0, base on A3 datasheet.

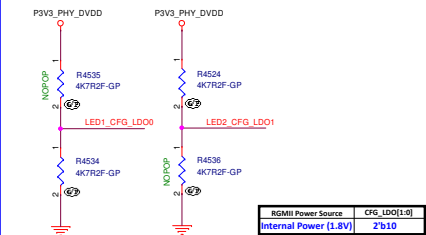
<Variant Name>

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Title BMC_STRAPS		
Size A3	Document Number 21X000	Rev SA
Date: Friday, May 28, 2021	Sheet 35	of 67

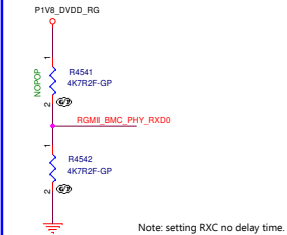
BMC_RGMII_PHY



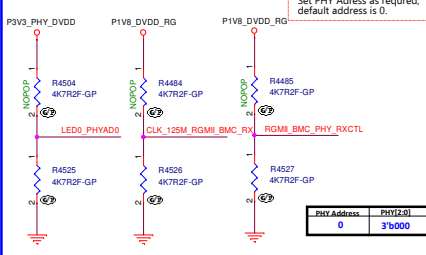
RGMII Voltage config.



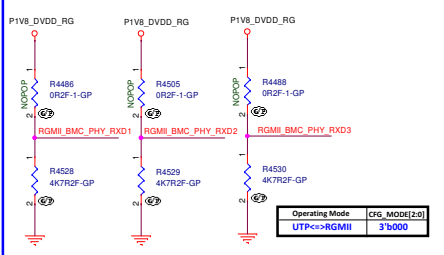
RGMII RXC Delay config.



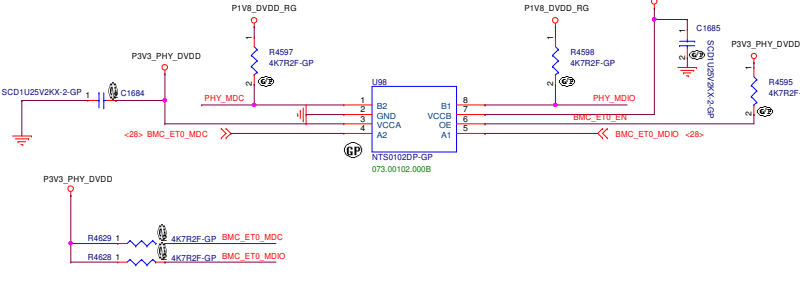
PHY ADDRESS CONFIG.



PHY CONFIG.



MDC/MDIO LEVEL SHIFTER



Front View

Left LED	Right LED
LED0	LED1
10M	10M
GREEN	AMBER

Note:
1.link up= bright
2.transmitting or receiving=blinking

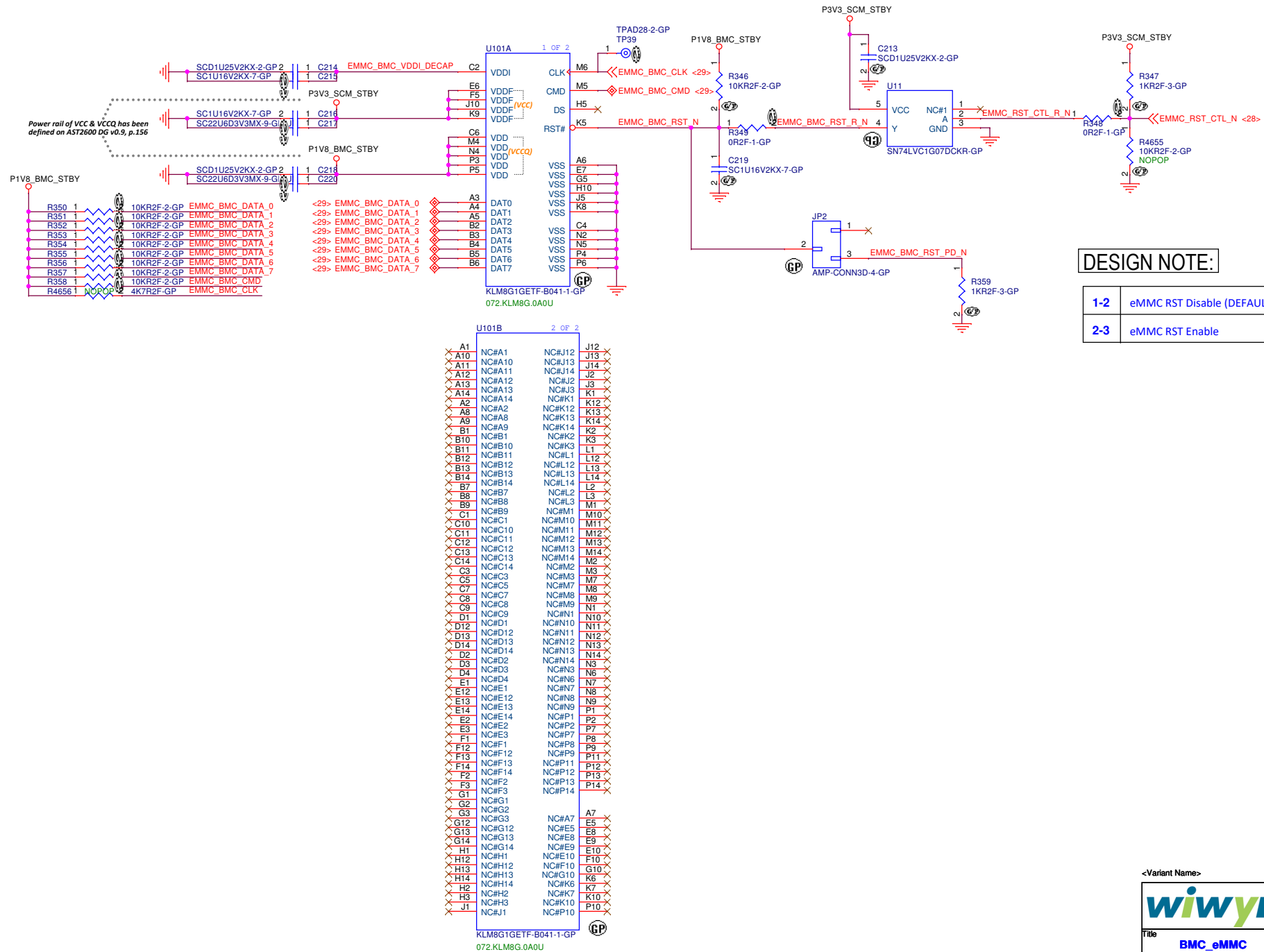
	Default status
LED0/PHYAD0	0
LED1/CFG_LD00	0
LED2/CFG_LD01	1

<Variant Name>



File	BMC_RGMII_PHY	Rev	SA
Size	21K000	Document Number	
Date	Friday, Mar 28, 2021	Sheet	36 of 57

BMC_eMMC



DESIGN NOTE:

1-2	eMMC RST Disable (DEFAULT)
2-3	eMMC RST Enable

<Variant Name>

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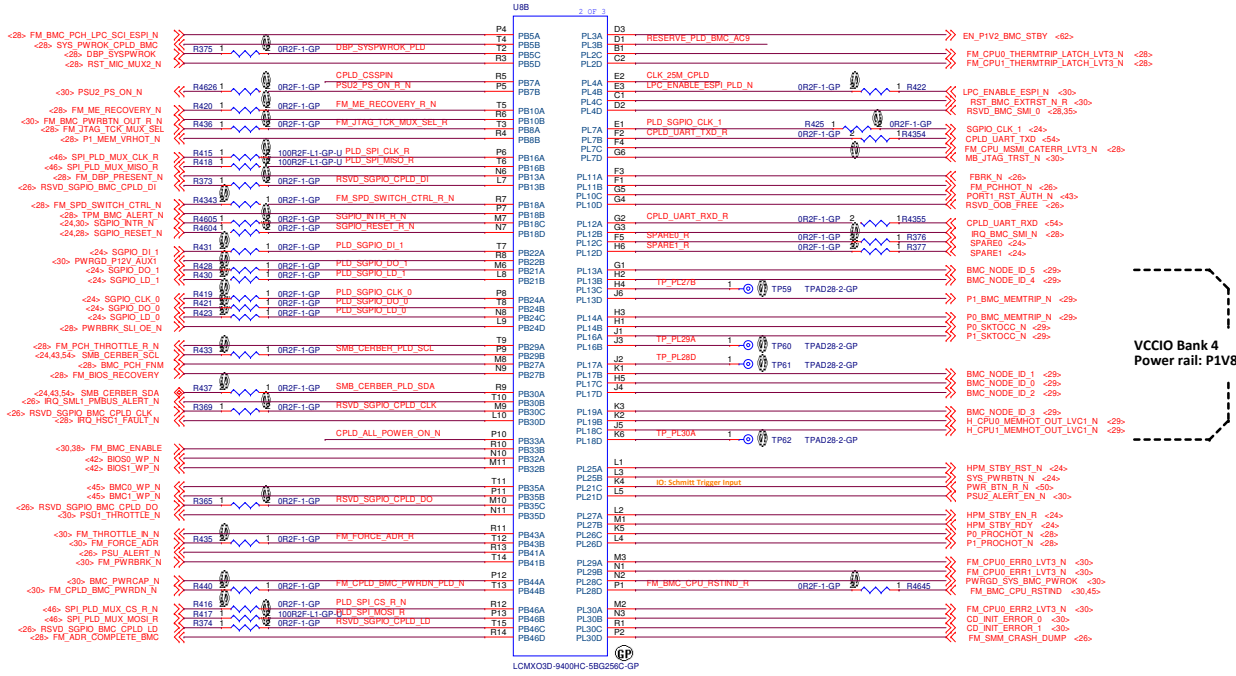
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Size A3	Document Number 21X000
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Date: Friday, May 28, 2021 Sheet 37 of 67

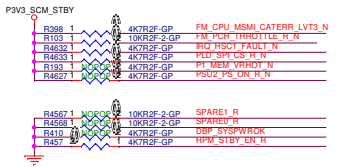
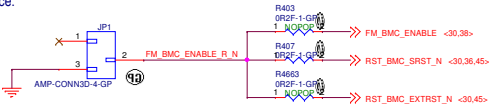
SA

CPLD (1/3)



DESIGN NOTE:
When jumper is installed on pins 2 and 3, SRST# on BMC is pulled low and all pins on BMC are set to high impedance.

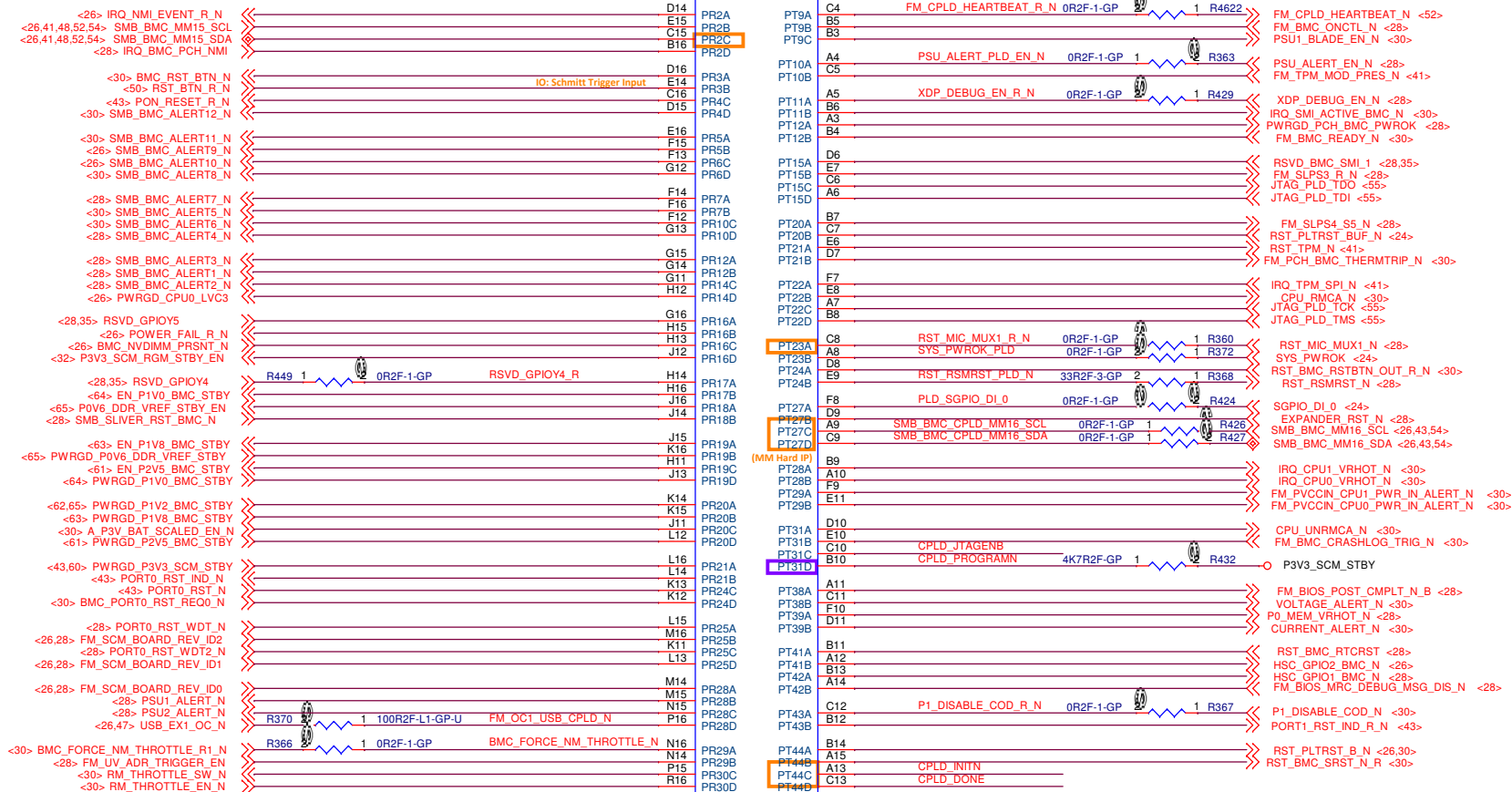
1-2	BMC enable (DEFAULT)
2-3	BMC disable



DESIGN NOTE:
1. NOPOP P01 MEM_VRHOT_N PU Res, because follow Lenovo design.
2. NOPOP SPARED1_R PD Res, because follow Lenovo design.
3. NOPOP PSU2_PS_ON_R_N PU Res, follow LENO
4. Put R10 BAT_GF in virtual EXP
5. *RESERVE_PLD_BMC_AC9* only connect to CPLD no BMC



CPLD (2/3)



LCMX03D-9400HC-5BG256C-GP

P3V3_SCM_STBY

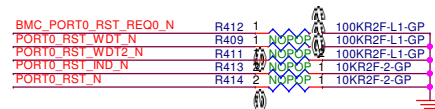
P3V3_SCM_STBY

DESIGN NOTE:

SMBUS ADDRESSES MM15 (SW): SMBUS ADDRESSES MM16 (HW):
0x60 (7-bit slave address) 0x40 (7-bit slave address)
0xC0 (8-bit write address) 0x80 (8-bit write address)
0xC1 (8-bit read address) 0x81 (8-bit read address)

DESIGN NOTE:

1."CPLD_ALL_POWER_ON_N" for SCM board all Power on (Force power on)



SMB_BMC_ALERT1_N	4K7R2F-GP	2	1	R627
SMB_BMC_ALERT2_N	4K7R2F-GP	1	1	R629
SMB_BMC_ALERT3_N	4K7R2F-GP	1	1	R631
SMB_BMC_ALERT4_N	4K7R2F-GP	1	1	R632
SMB_BMC_ALERT5_N	4K7R2F-GP	1	1	R633
SMB_BMC_ALERT6_N	4K7R2F-GP	1	1	R634
SMB_BMC_ALERT7_N	4K7R2F-GP	1	1	R635
SMB_BMC_ALERT8_N	4K7R2F-GP	1	1	R636
SMB_BMC_ALERT9_N	4K7R2F-GP	1	1	R637
SMB_BMC_ALERT10_N	4K7R2F-GP	1	1	R638
SMB_BMC_ALERT11_N	4K7R2F-GP	1	1	R4572
SMB_BMC_ALERT12_N	4K7R2F-GP	1	1	R4573
FM_THROTTLE_SW_N	4K7R2F-GP	1	1	R391
BMC_FORCE_NM_THROTTLE_N	4K7R2F-GP	1	1	R393
PSU2_ALERT_N	4K7R2F-GP	1	1	R378
PSU1_ALERT_N	4K7R2F-GP	1	1	R379
PORT0_RST_WDT_N	4K7R2F-GP	1	1	R380
PORT0_RST_WDT2_N	4K7R2F-GP	1	1	R381
PORT0_RST_IND_N	4K7R2F-GP	1	1	R383
PORT0_RST_WDT2_N	4K7R2F-GP	1	1	R384
POWER_FAIL_R_N	4K7R2F-GP	1	1	R71
BMC_RST_BTN_N	4K7R2F-GP	1	1	R446
RST_RSMRST_PLD_N	4K7R2F-GP	1	1	R4638
CPLD_DONE	10KR2F-2-GP2	1	1	R4642
CPLD_INITN	10KR2F-2-GP2	1	1	R4643
CPLD_JTAGENB	4K7R2F-GP	1	1	R4644
BMC_PORT0_RST_REQ0_N	4K7R2F-GP	1	1	R382
PWRGD_CPU0_LVC3	4K7R2F-GP	1	1	R75
HSC_GPIO2_BMC_N	4K7R2F-GP	1	1	R4637
HSC_GPIO2_BMC_N	4K7R2F-GP	1	1	R67
P0_MEM_VRHOT_N	4K7R2F-GP	1	1	R194

<Variant Name>

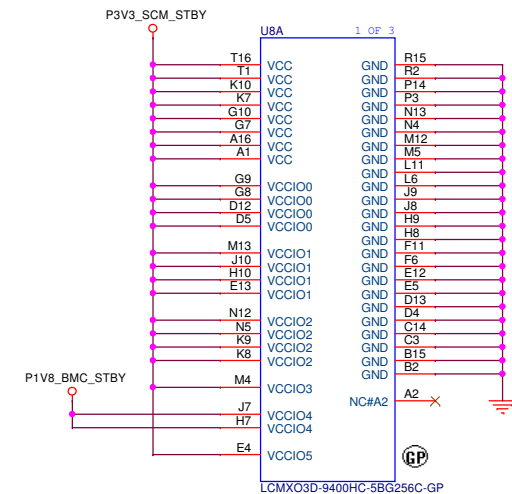
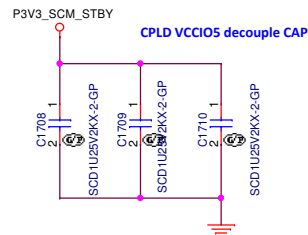
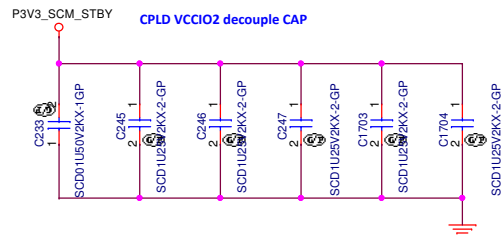
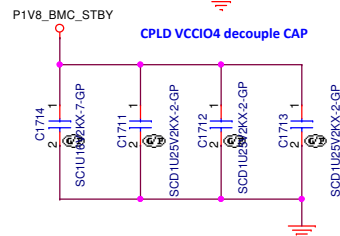
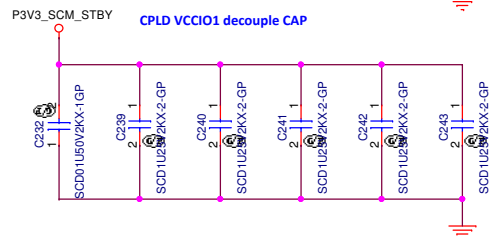
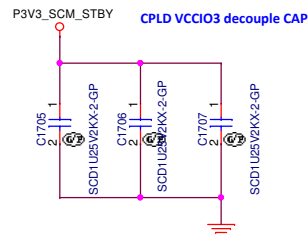
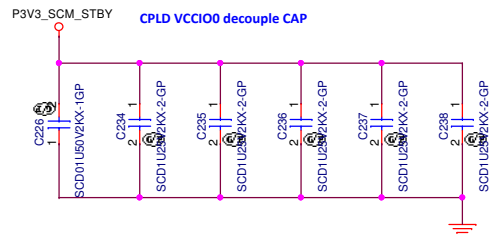
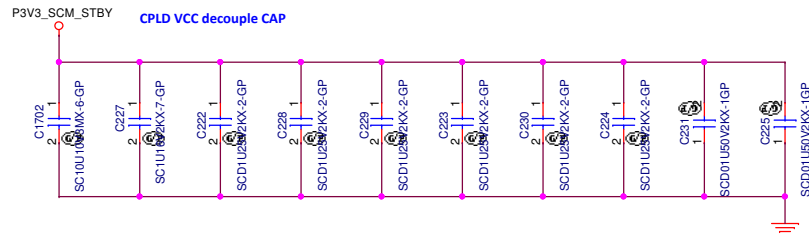
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Title: **CPLD**

Size: A3 Document Number: **21X000** Rev: SA

Date: Wednesday, June 16, 2021 Sheet: 39 of 67

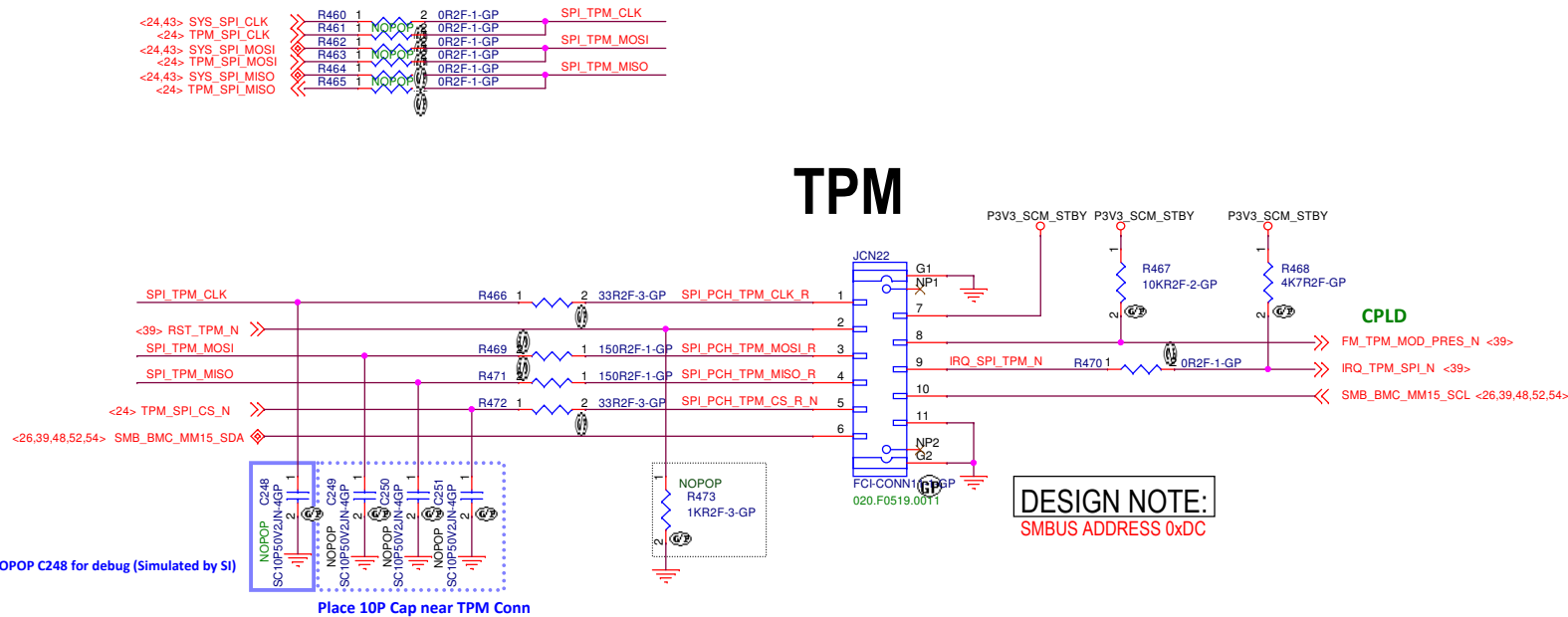
CPLD (3/3)



DESIGN NOTE:

Place these CAP close to CPLD

TPM_CONNECTOR

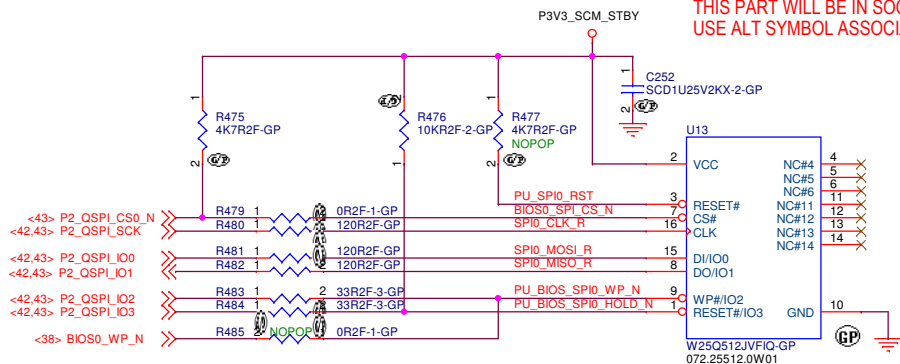


Remove P3V3 power rail reserved for other platform.

BIOS 0

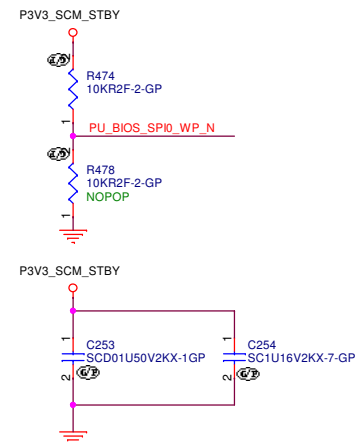
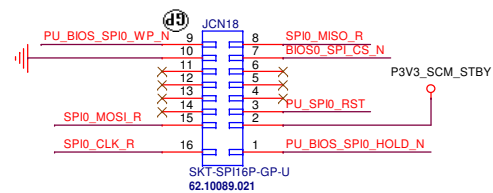
CAD NOTE:

THIS PART WILL BE IN SOCKET, PLEASE
USE ALT SYMBOL ASSOCIATED WITH PART FOOTPRINT.

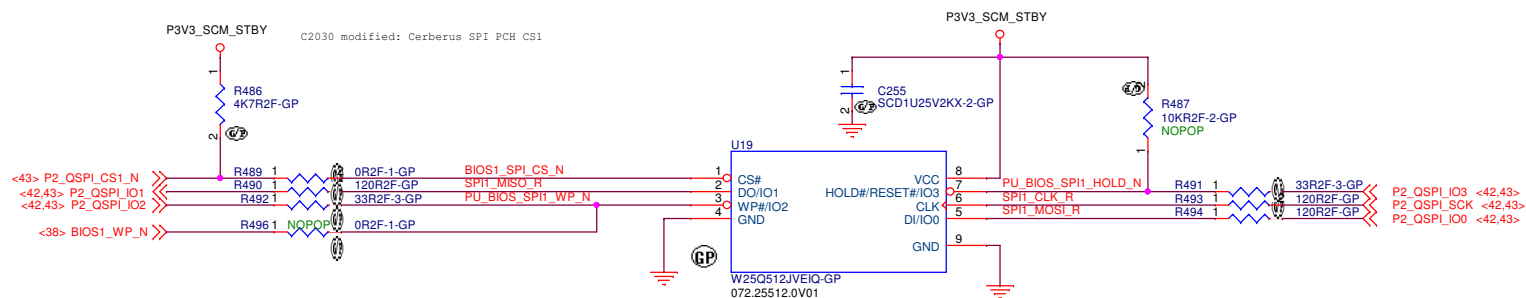


DIP
2nd:

Package: SOP-16

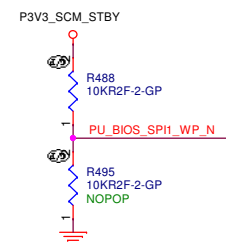


BIOS 1



SMT
2nd:

Package: WSON



<Variant Name>



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Title	BIOS_FLASH
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Size A3	Document Number 21X000
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Rev	SA
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Date: Friday, May 28, 2021

Sheet 42 of 67

		Wiyynn Incorporated 8F, No 93, Xindai 5th Rd., Xizhi Dist. New Taipei, Taiwan http://www.wiyynn.com	
Cerberus Connector			
Title	Document Number		
Size A2	21X000		
Date:	Friday, May 28, 2021	Sheet	43 of 67
		Rev	SA

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<Variant Name>



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Size

A3

Document Number

21X000

Rev

SA

Date:

Friday, May 28, 2021

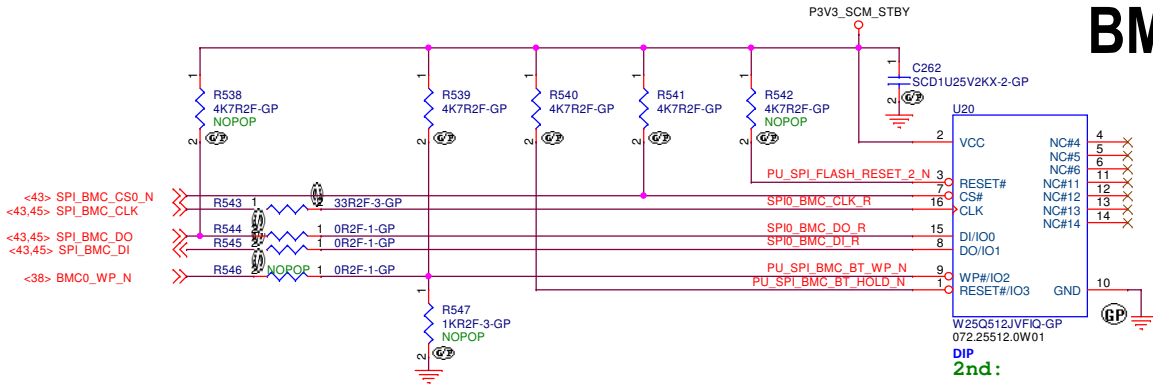
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44

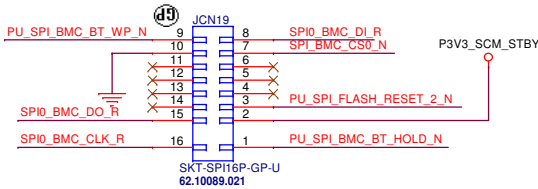
of

67

BMC FLASH



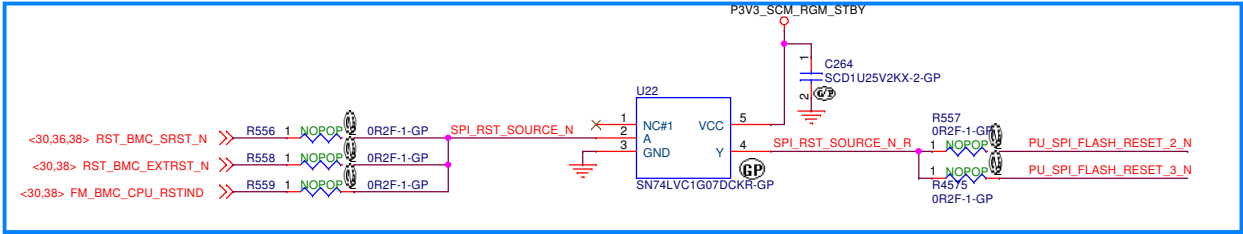
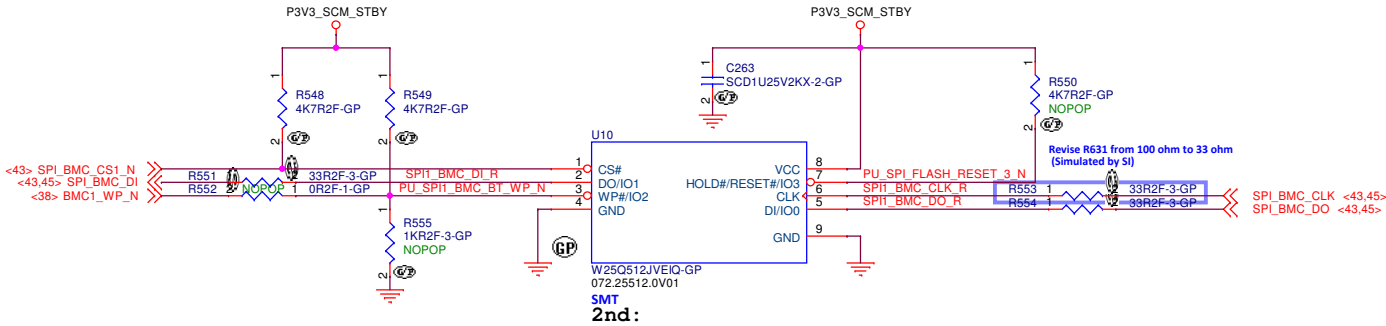
BMC 0



CAD NOTE:

PLACE CAPS NEAR VCC PIN ON W25Q256 COMPONENT

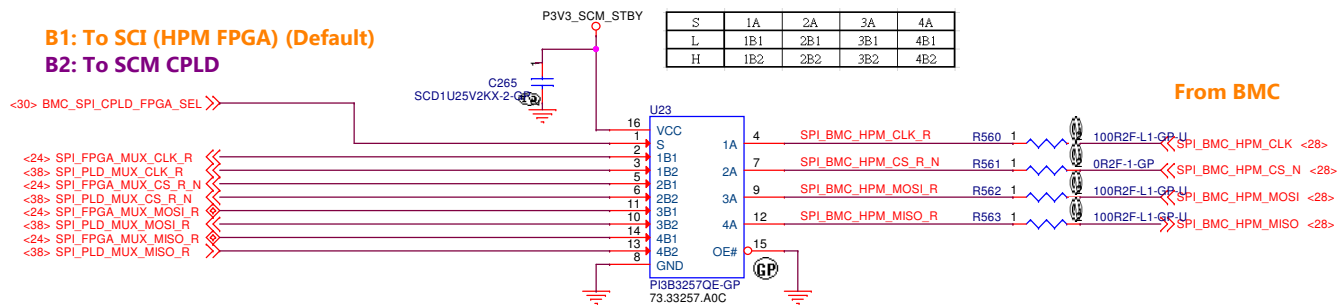
BMC 1



According to ASPEED design guide v09 p.147, EVB v14a p.15.
Add buffer for the connection between RSTIND#/SRST#/EXTRST and SPI_RST#.

<Variant Name>

HPM CPLD SPI MUX



DESIGN NOTE:

Res tuning after SPI signal simulation.

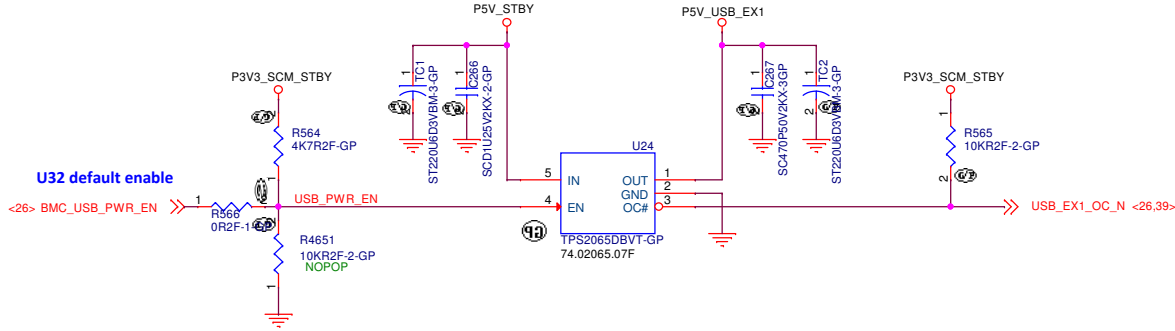
R560	100 ohm	R554	33 ohm
R415	100 ohm	R510	75 ohm
R562	100 ohm	R508	75 ohm
R563	100 ohm	R551	33 ohm
R417	100 ohm	R519	75 ohm
R418	100 ohm	R521	75 ohm
R502	120 ohm	R524	33 ohm
R504	75 ohm	R529	33 ohm
R505	75 ohm		

<Variant Name>

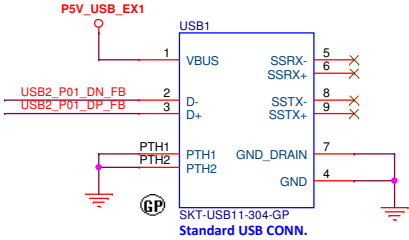
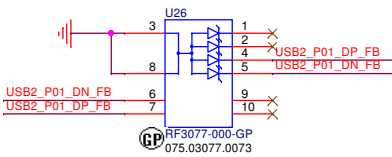
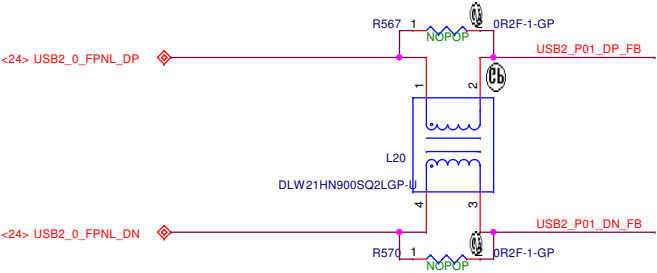
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Title HPM CPLD SPI MUX			
Size A3	Document Number 21X000		Rev SA
Date:	Friday, May 28, 2021	Sheet 46	of 67

USB

USB Power Delivery



✳ Revise the port according to layout routing



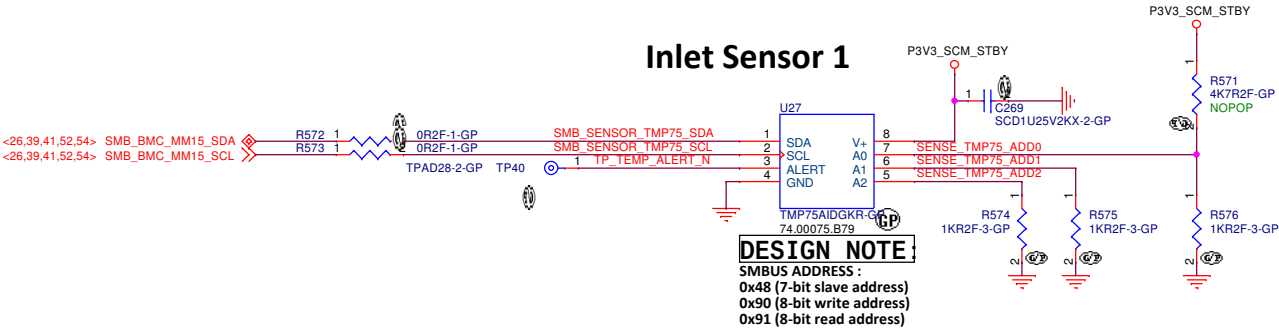
DESIGN NOTE:

1. Add 220uF TC1 to resolve USB OC issue
2. FPNL USB connector will remove in production.

<Variant Name>

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Title		USB	
Size	Document Number	Rev	
A3	21X000	SA	
Date:	Friday, May 28, 2021	Sheet	47 of 67

TEMPERATURE_SENSORS



Address table for TPM75

A2	A1	A0	ADDR
0	0	0	90
0	0	1	92

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<Variant Name>



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Size

A3

Document Number

21X000

Rev

SA

Date:

Friday, May 28, 2021

Sheet

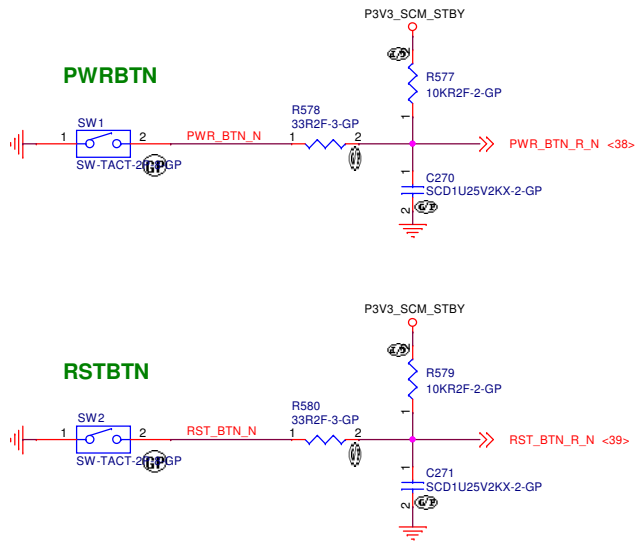
49

of

67

POWER_RESET_BUTTON

Put buffer and logic circuit into SCM CPLD
The PWRBTN/RSTBTN circuit logic is follwo C2080.



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<Variant Name>



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Size

A3

Document Number

21X000

Rev

SA

Date:

Friday, May 28, 2021

Sheet

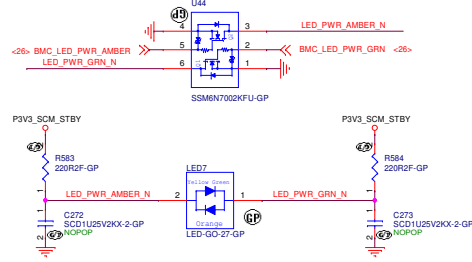
51

of

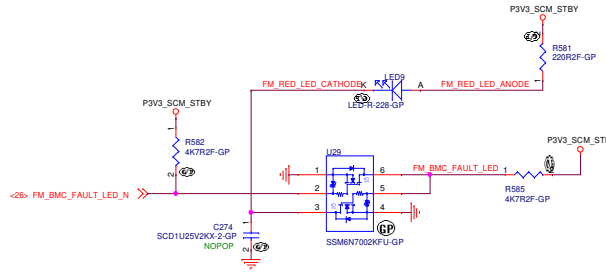
67

FUNCTION LED _ FRU

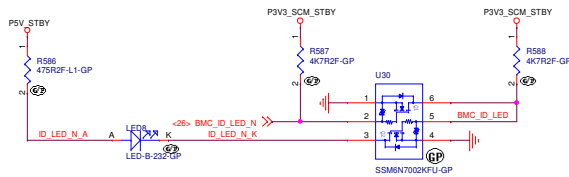
Power state LED



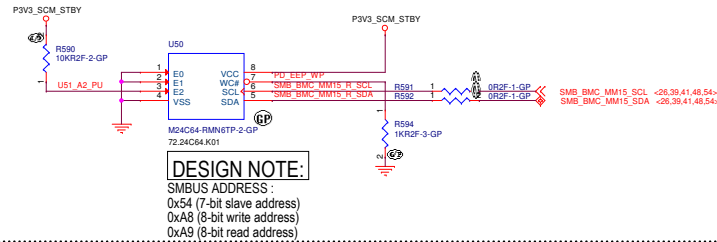
BLADE ATTENTION LED



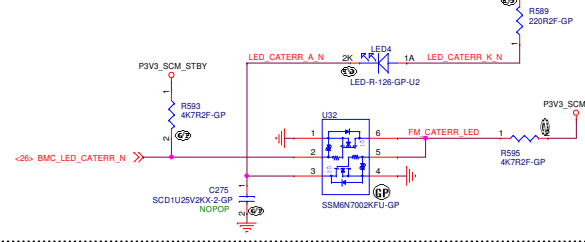
ID LED



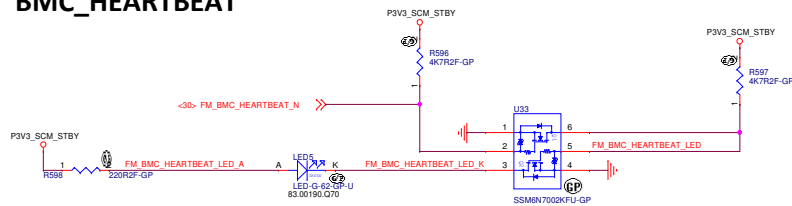
FRU



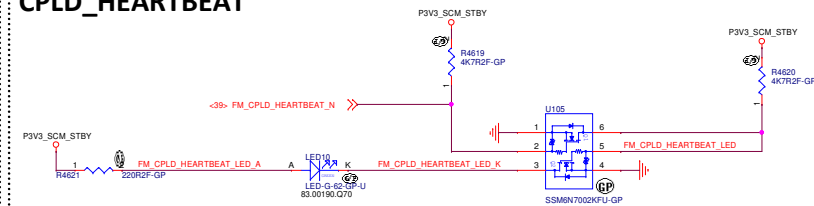
CATERR LED



BMC_HEARTBEAT



CPLD_HEARTBEAT



<Variant Name>

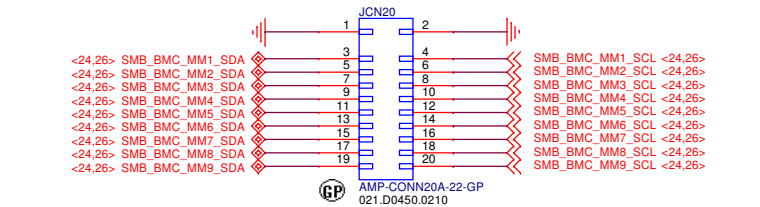
wiwynn Wiwynn Incorporated
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File
Size A2
Document Number
21X000
Date: Wednesday, June 18, 2021
Sheet 52 of 57
Rev SA

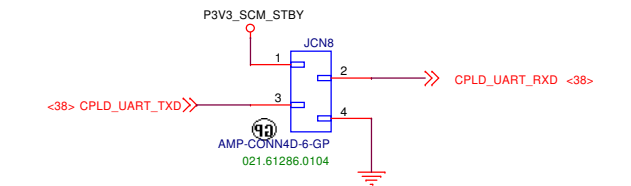
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I2C_I3C_UART_Header

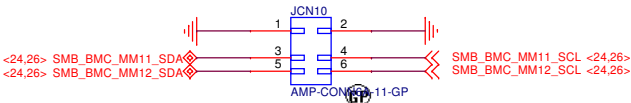
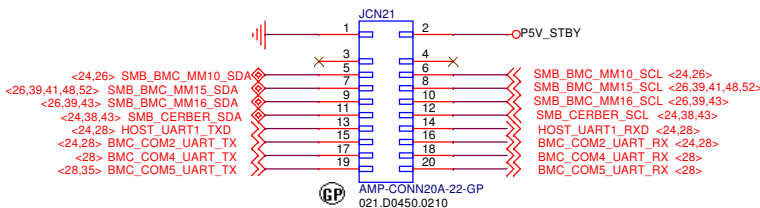
I2C Header



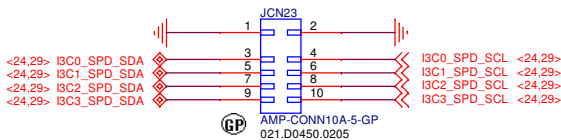
CPLD UART Debug Header



UART / I2C Header



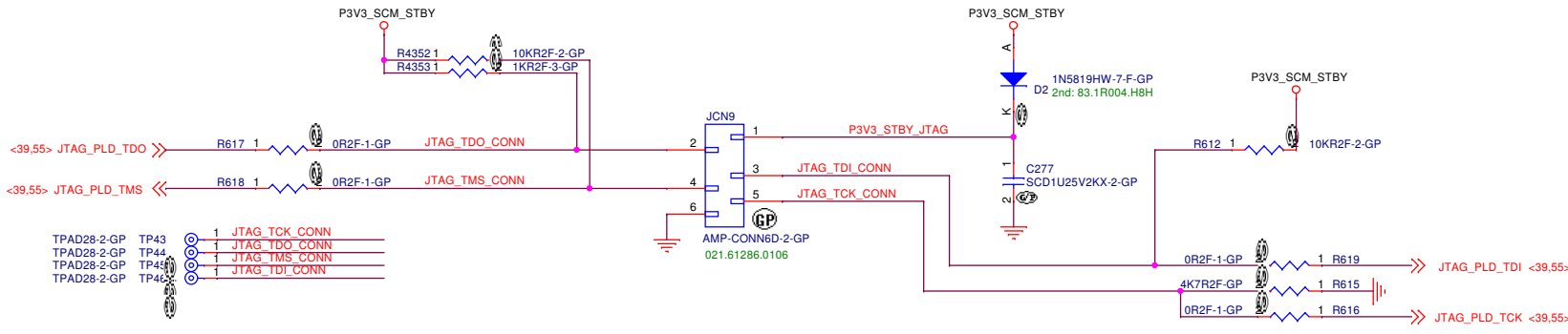
I3C header



PLD_JTAG_HEADERS

To CPLD

From BMC



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Title			
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Size	Document Number		Rev
A3	21X000		SA
Date:	Friday, May 28, 2021		Sheet 56 of 67

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<Variant Name>



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A3

Document Number

21X000

Rev

SA

Date:

Friday, May 28, 2021

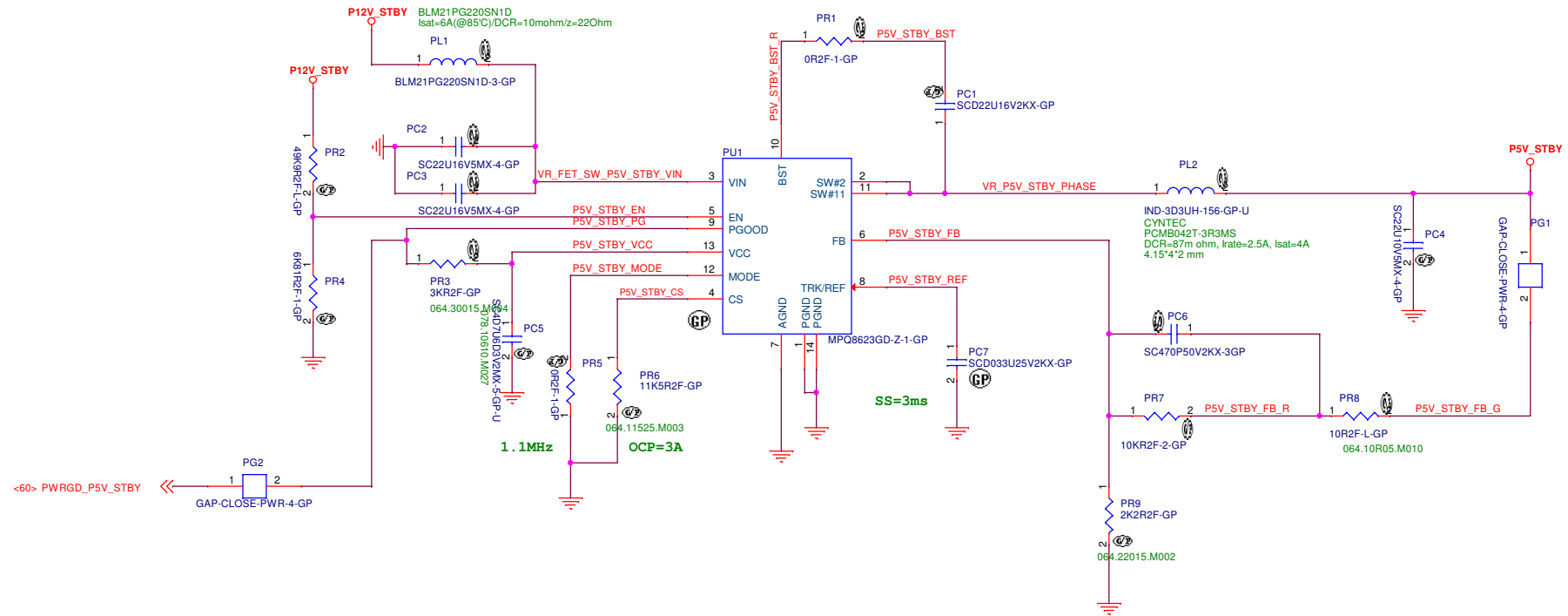
Sheet

58

of

67

Power rail : P5V_STBY
Controller : MPQ8623
TDC :A
Imax :2A
OCP: 3A



<Variant Name>



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Title **P5V_STBY**

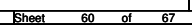
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Date: Friday, May 28, 2021

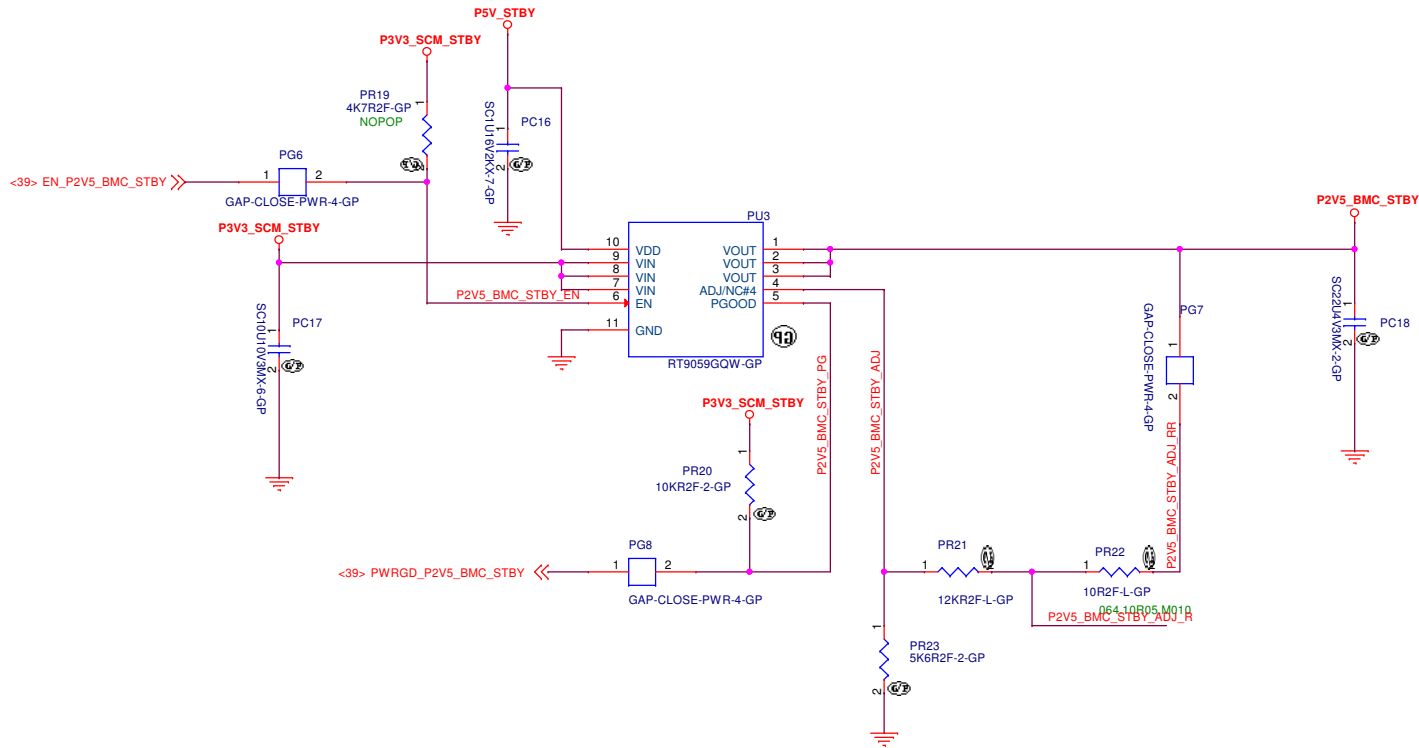
Rev
SA

Sheet 59 of 67

OCP=8.5A



Power rail : P2V5_BMC_STBY
Controller : RT9059GQW
Itdc=A, I_{max}=1.6A
OCP=3.6A (fixed)

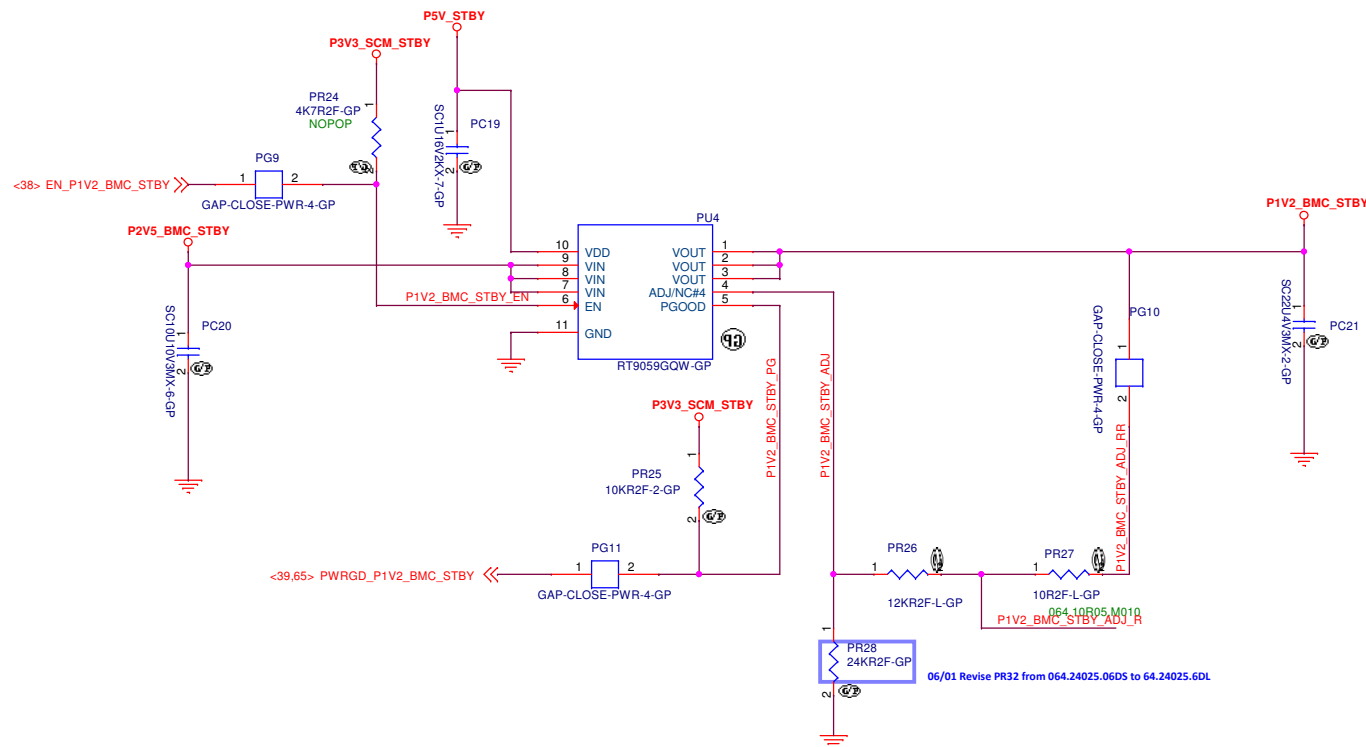


<Variant Name>

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Title		P2V5_BMC_STBY	
Size	Document Number	Rev	
A3	21X000	SA	
Date:	Friday, May 28, 2021	Sheet	61 of 67

Power rail : P1V2_BMC_STBY
Controller : RT9059GQW
Imax=1.3A
OCP=3.6A (fixed)

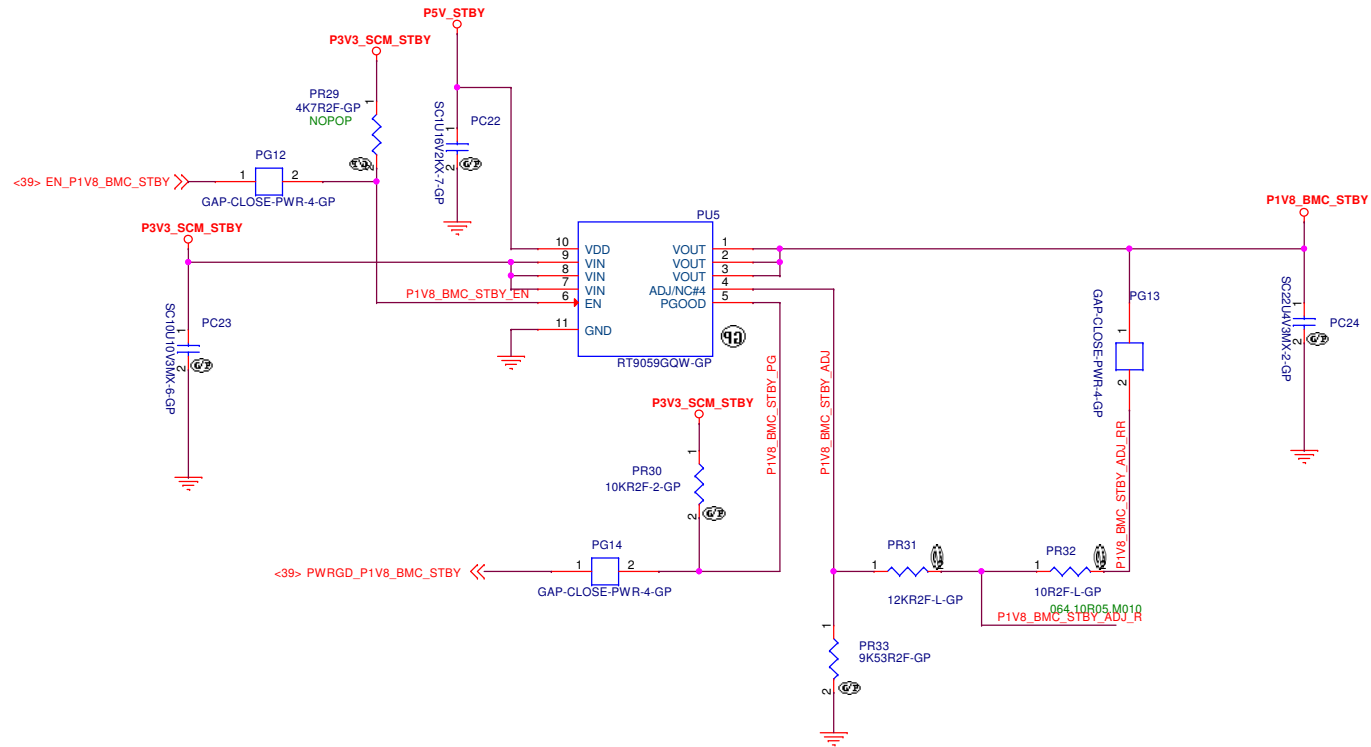
20200601 Update



<Variant Name>

wiwynn		Wiwynn Incorporated 8F, No.90, Xintai 5th Rd., Xizhi Dist, New Taipei, Taiwan http://www.wiwynn.com	
Title		P1V2_BMC_STBY	
Size	Document Number	Rev	
A3	21X000	SA	
Date:	Friday, May 28, 2021	Sheet	62 of 67

Power rail : P1V8_BMC_STBY
Controller : RT9059GQW
Imax=0.64A
OCP=3.6A (fixed)



<Variant Name>

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Title	P1V8_BMC_STBY
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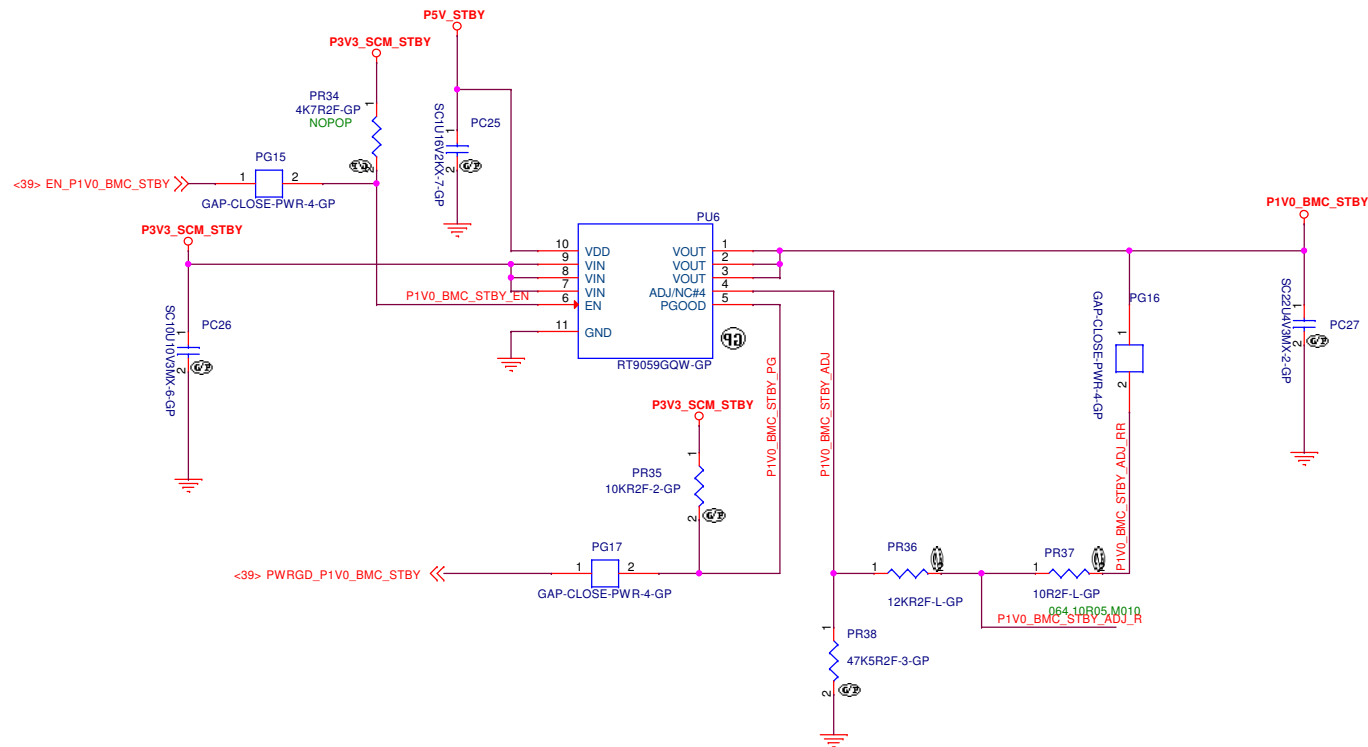
Size A3	Document Number 21X000
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Rev	
SA	

Date: Friday, May 28, 2021

Sheet 63 of 67

Power rail : P1V0_BMC_STBY
Controller : RT9059GQW
I_{ldc}=A, I_{max}=0.7A
OCP=3.6A (fixed)



<Variant Name>

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Title	P1V0_BMC_STBY
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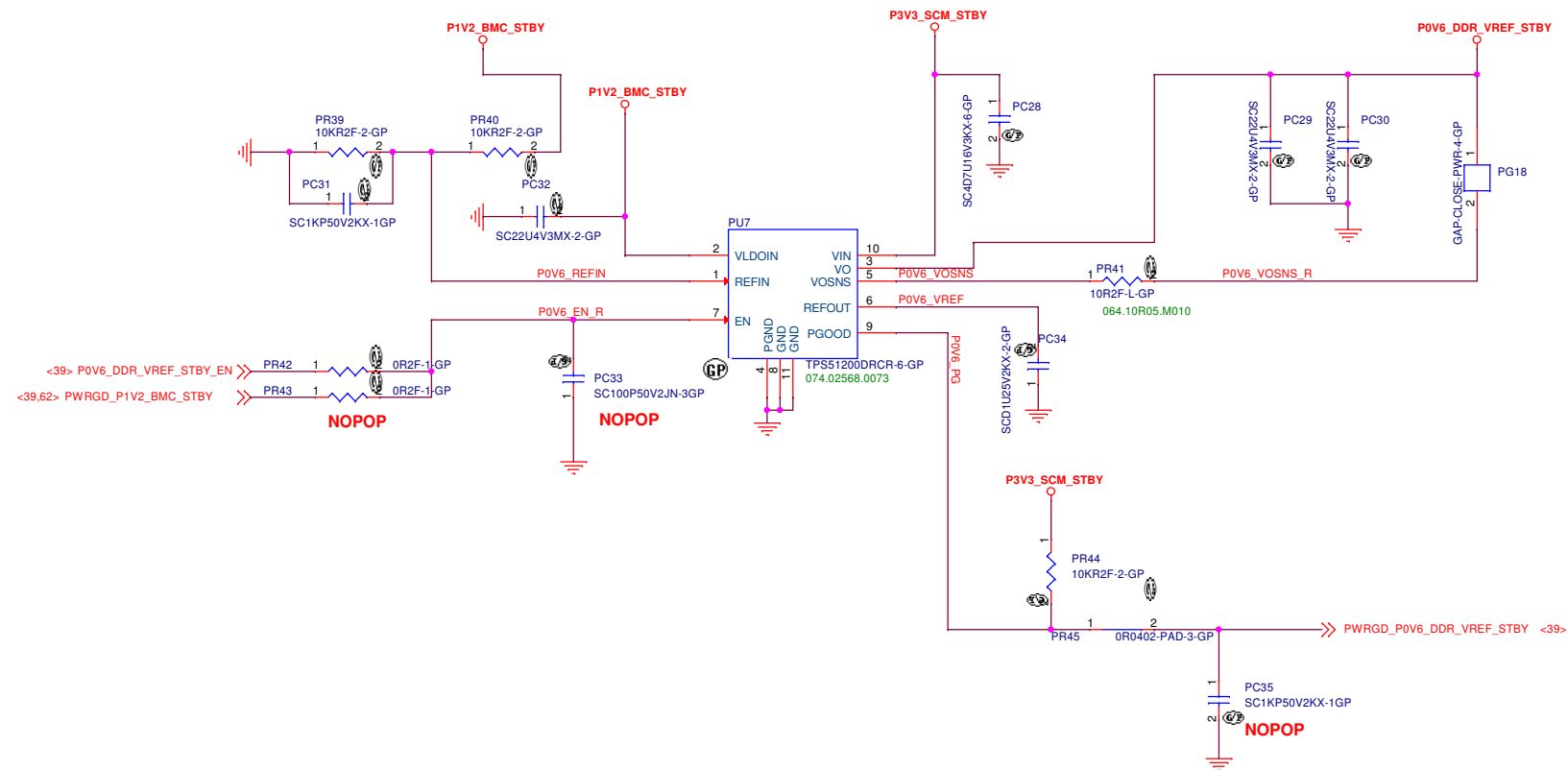
Size A3	Document Number 21X000
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Rev	
SA	

Date: Friday, May 28, 2021

Sheet 64 of 67

Power rail : P0V6_DDR_VREF_STBY
Controller : RT2568GQW(2)
Vout=0.6V
I_{max}=0.33A
OCP=4.5A



<Variant Name>



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Title	P0V6_DDR_VREF_STBY
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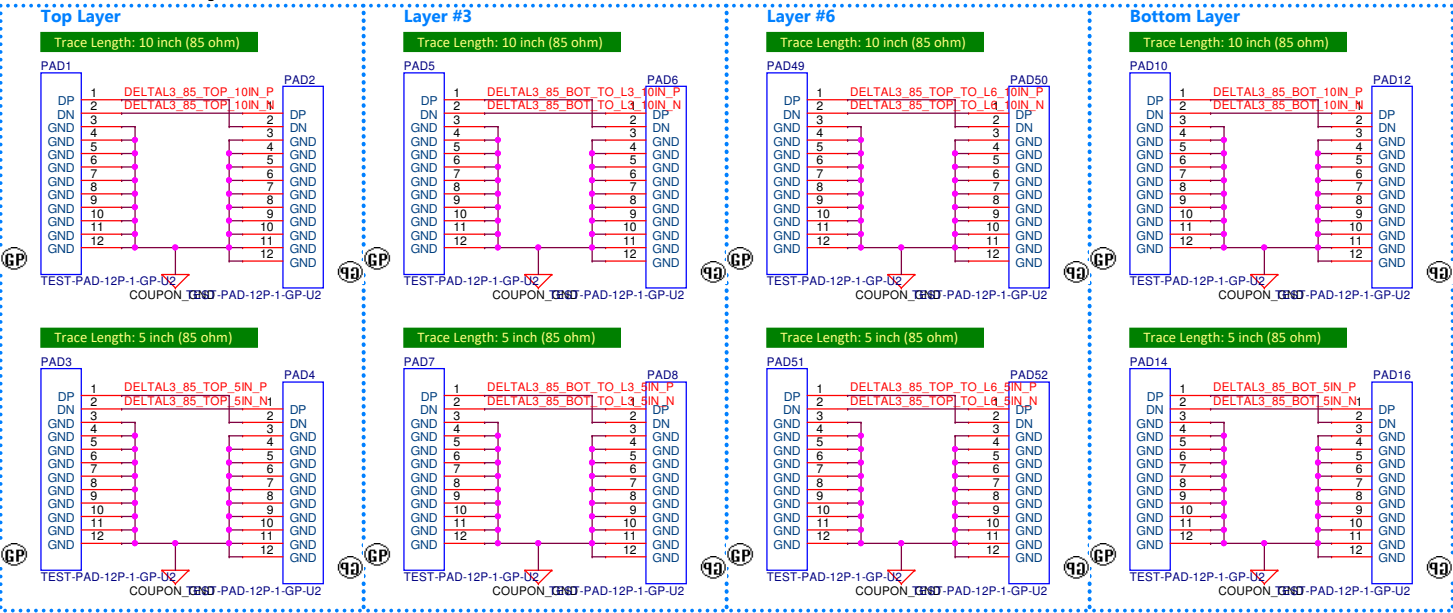
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Rev	
SA	

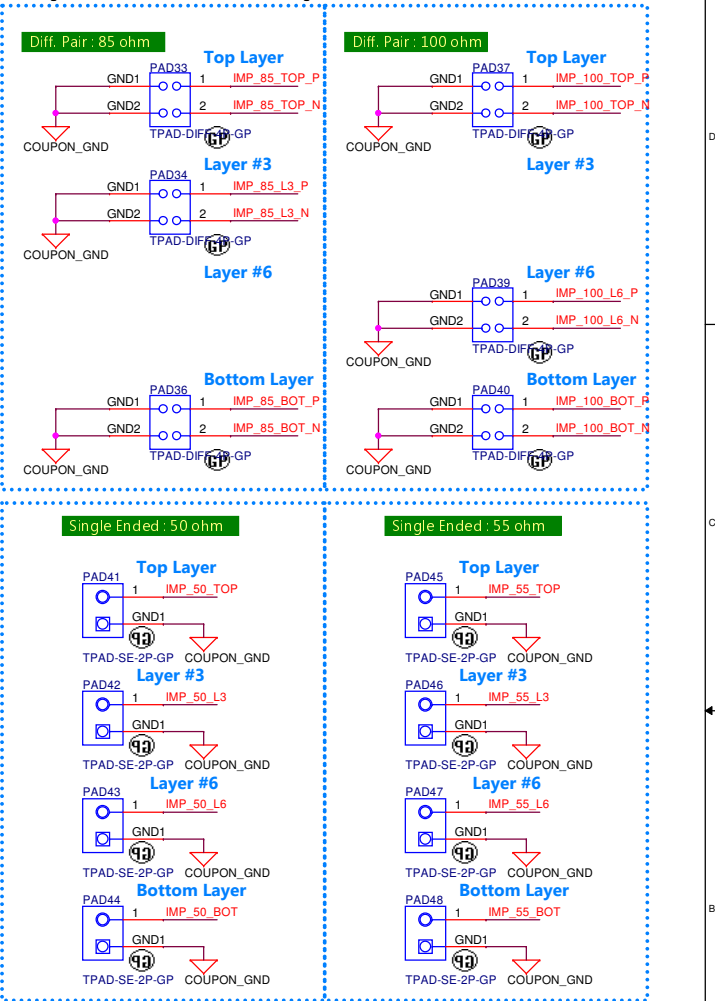
Date: Friday, May 28, 2021

Sheet 65 of 67

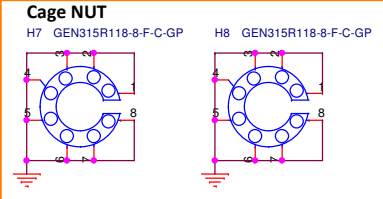
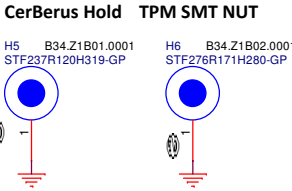
Delta-L Coupon



Impedance Coupon



Mechanical_Hole



<Variant Name>

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Title **HOLE**

Size A3 Document Number **21X000** Rev SA

Date: Friday, May 28, 2021 Sheet 66 of 67

History